Analysis of tropical cyclone intensity variability from real-time forecast

Chanh Kieu, Indiana University Bloomington

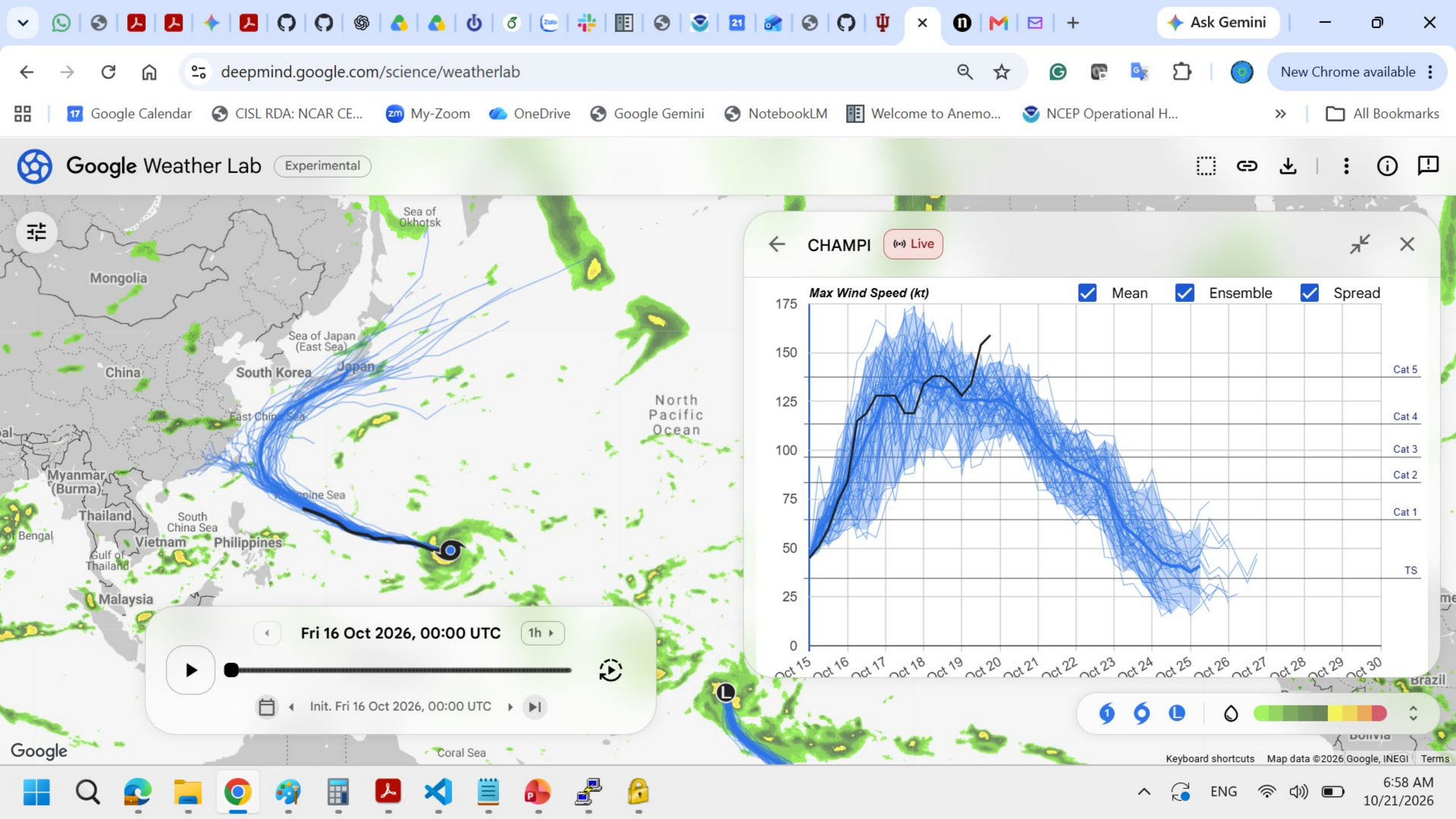
Objectives

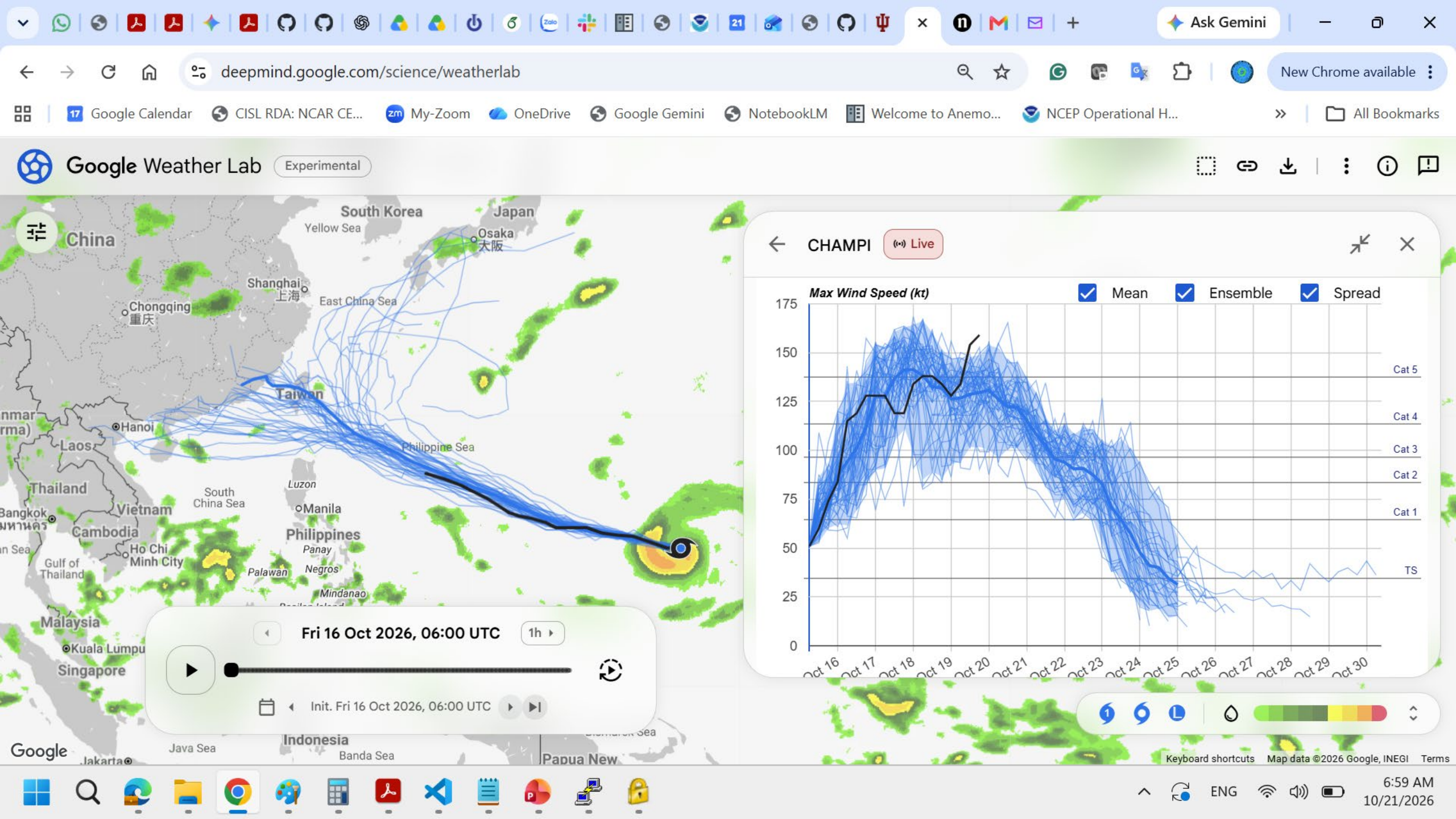
- Try to answer the question: Does TC dynamics possess any chaotic dynamics that can be manifested in or captured from real TC cases;
- Verify the recently proposed low-order TC intensity chaotic model with real TC cases

Data source

- Use the operational Google Weather next 3 TC forecast https://deepmind.google.com/science/weatherlab
- Focus on 2 recent examples of extreme cases Hurricane Polo (2026) and Super-Typhoon Chambi (2026)
- Use all available ensemble version from Google WeatherNext
- Analyze the error growth for different initial intensity values to distinguish the “transient orbit” versus “central orbit” for chaotic dynamics (Lorenz 1963, 1969)

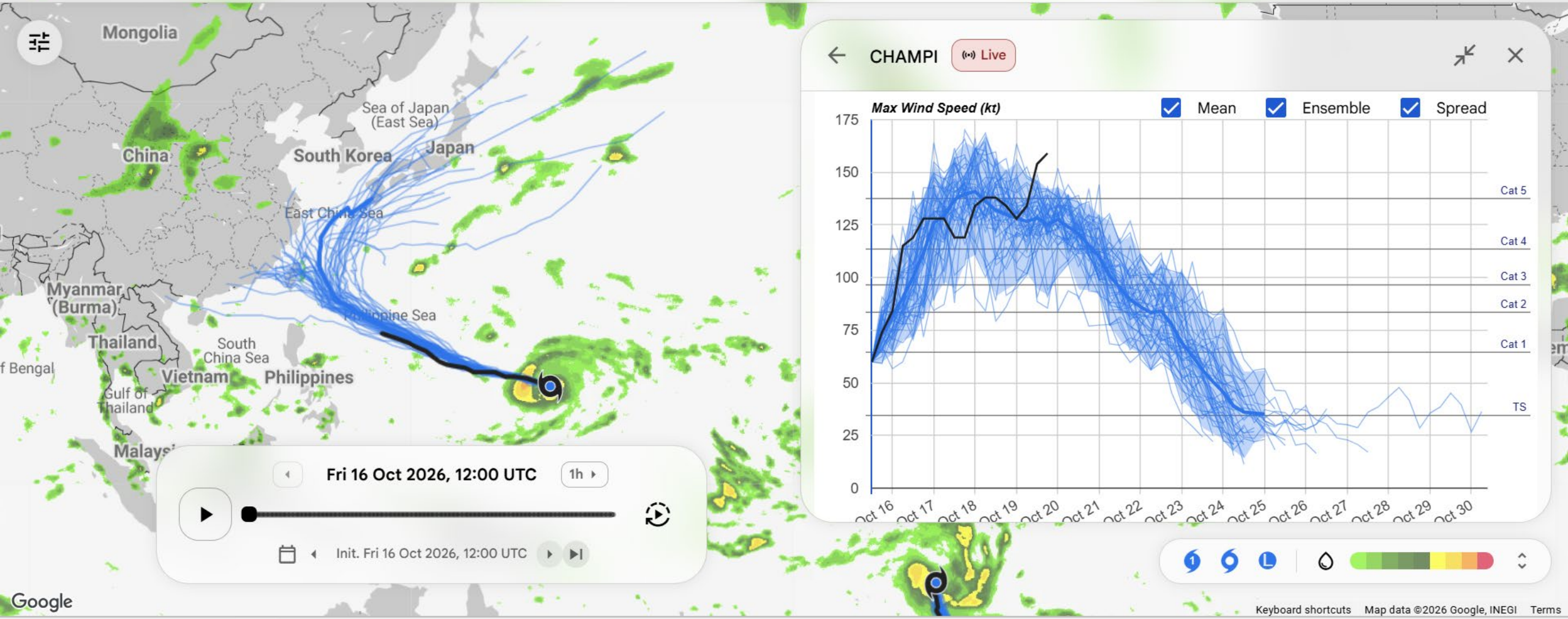
Super-typhoon Chambi (2026)


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Sea of Okhotsk
Mongolia
Sea of Japan (East Sea)
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South Korea
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East China Sea
North Pacific Ocean
Myanmar (Burma)
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Malaysia
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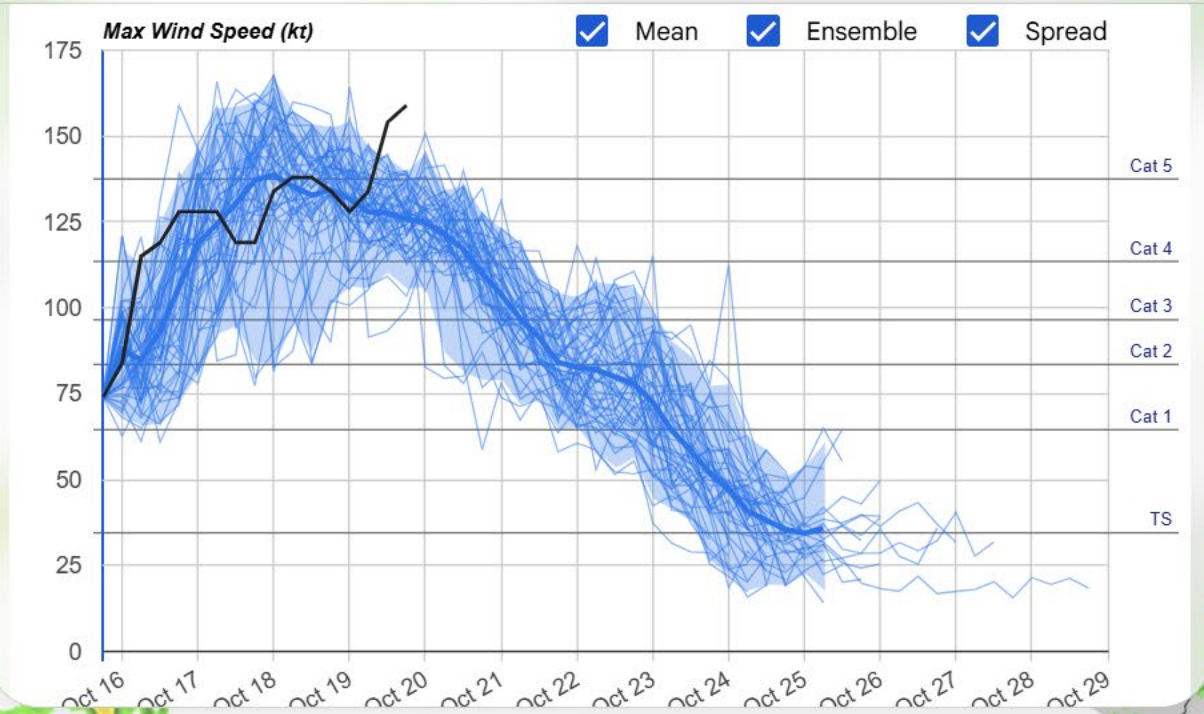


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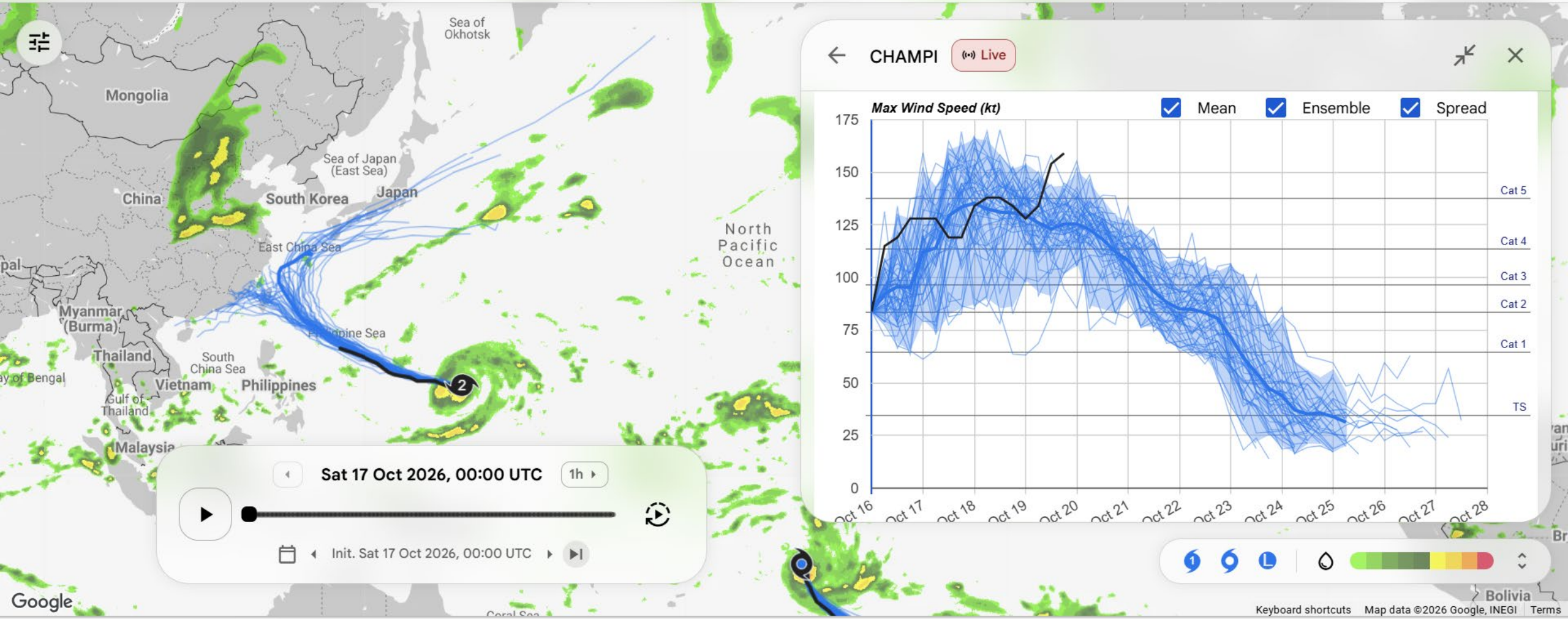
Mongolia China South Korea Japan Sea of Japan (East Sea) East China Sea Philippine Sea South China Sea Vietnam Philippines Thailand Myanmar (Burma) Gulf of Thailand Okhotsk

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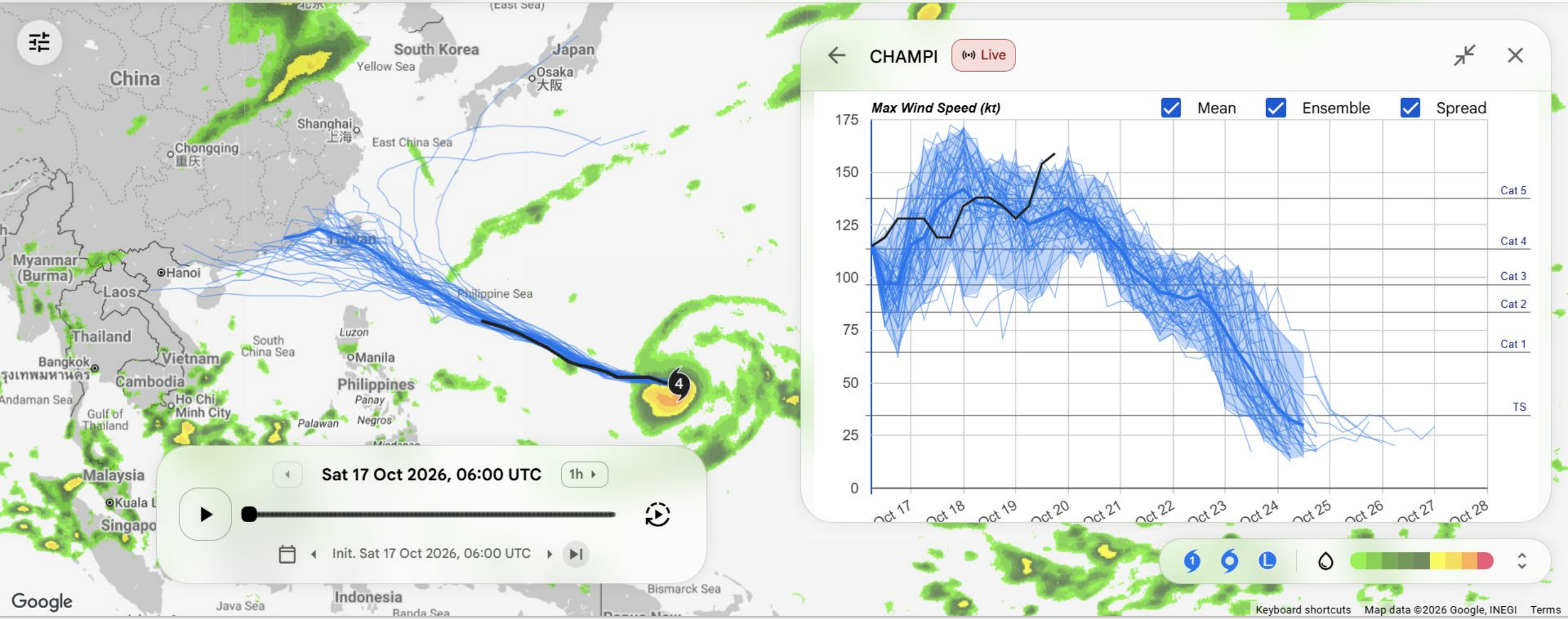
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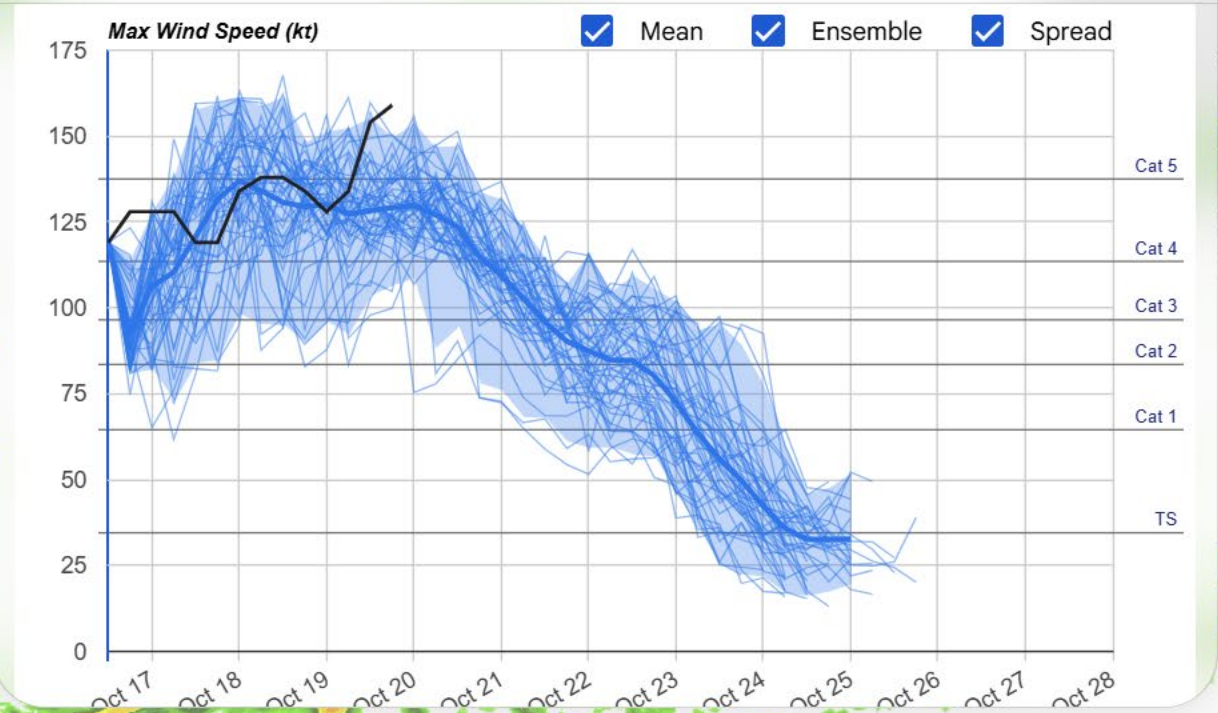
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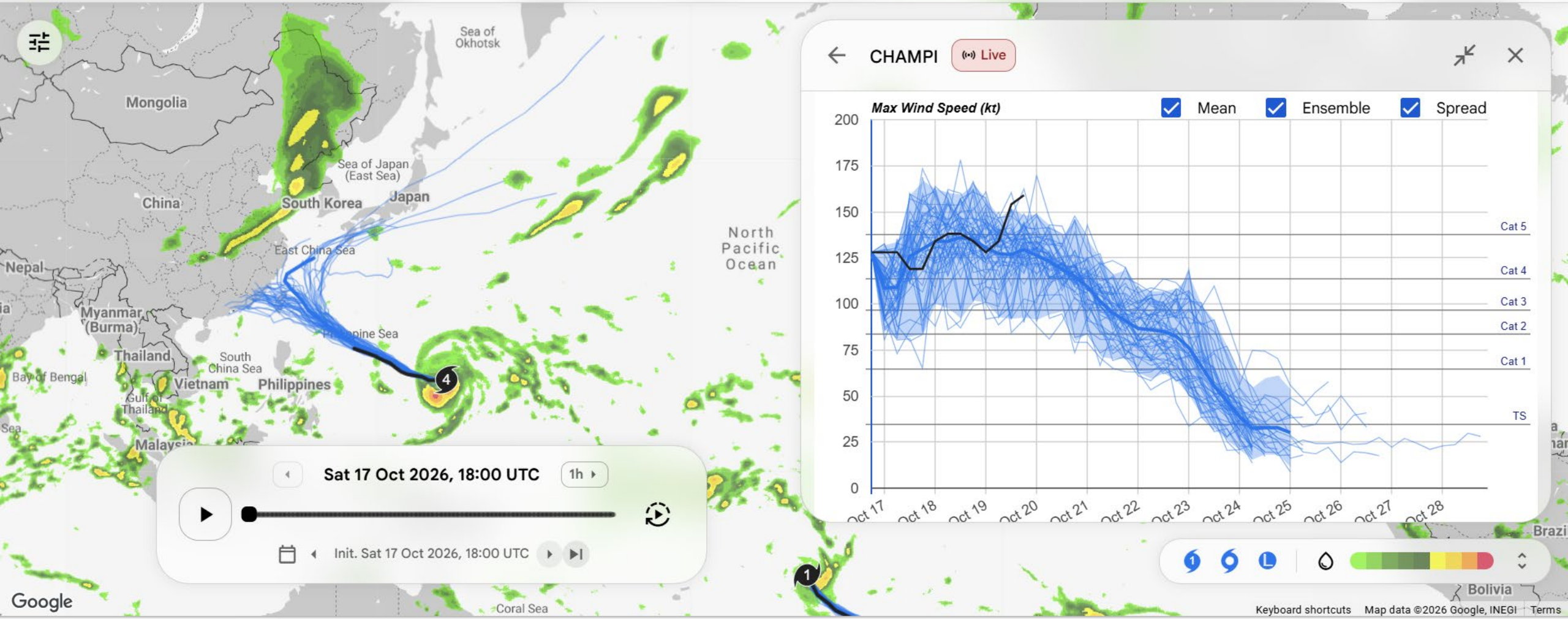
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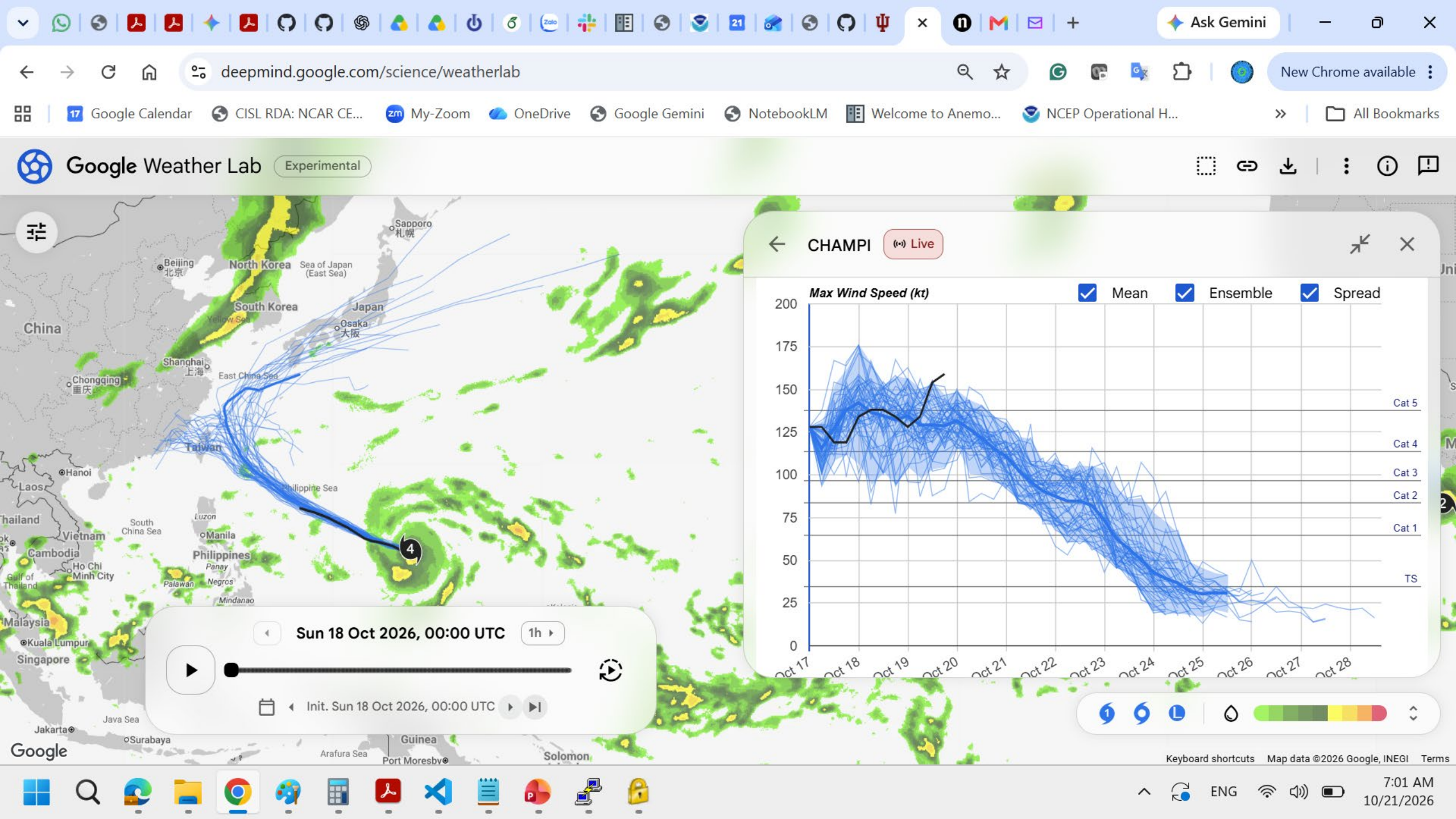


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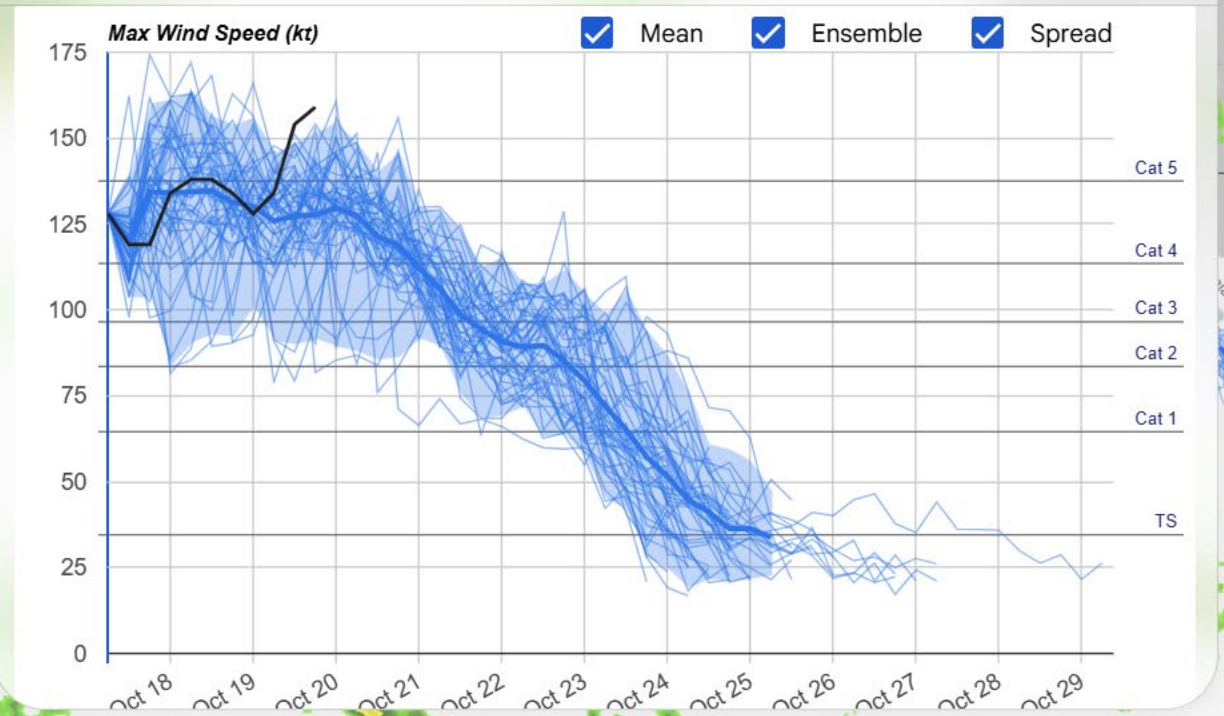
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Yellow Sea
Japan
Osaka
大阪
China
Shanghai
上海
East China Sea
Chongqing
重庆
Taiwan
Hanoi
Laos
Philippine Sea
South
China Sea
Luzon
Manila
Philippines
Panay
Palawan
Negros
Mindanao
Vietnam
Cambodia
Ho Chi
Minh City
Gulf of
Thailand
Malaysia
Kuala Lumpur
Singapore
Java Sea
Jakarta
Surabaya
Arafura Sea
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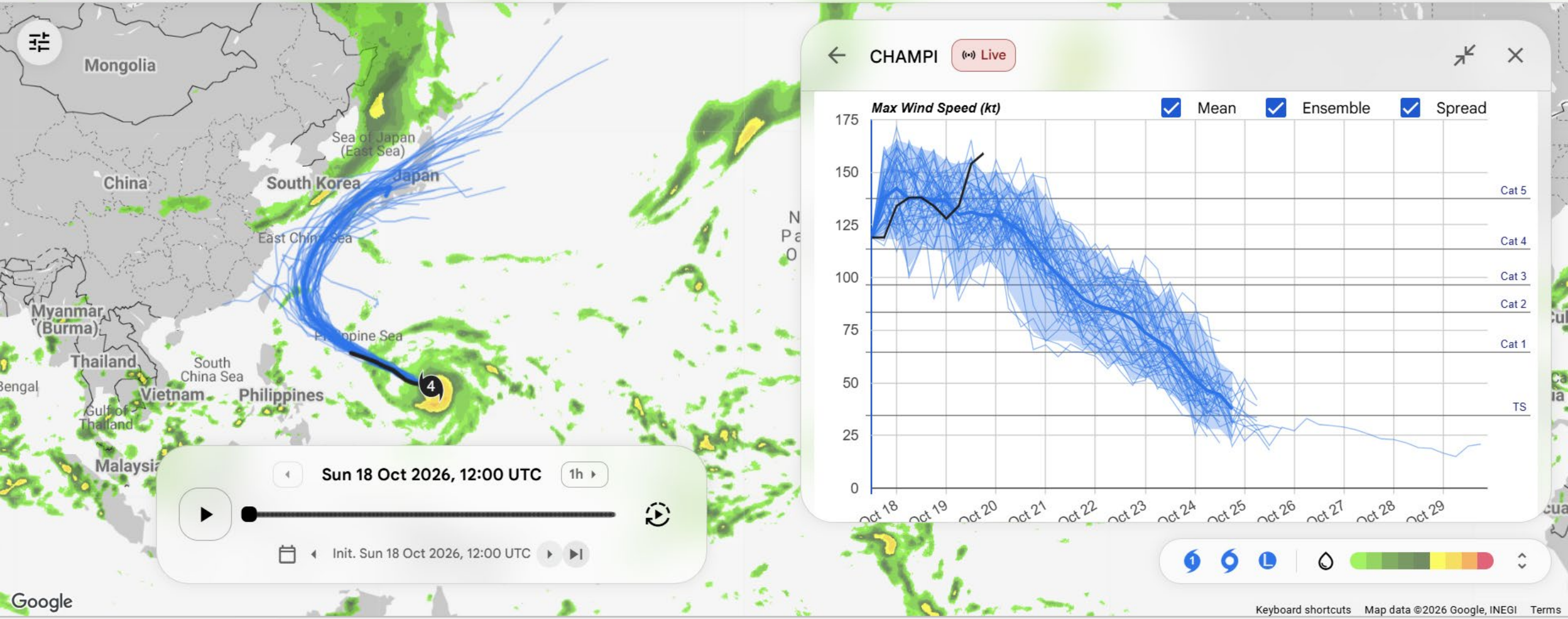
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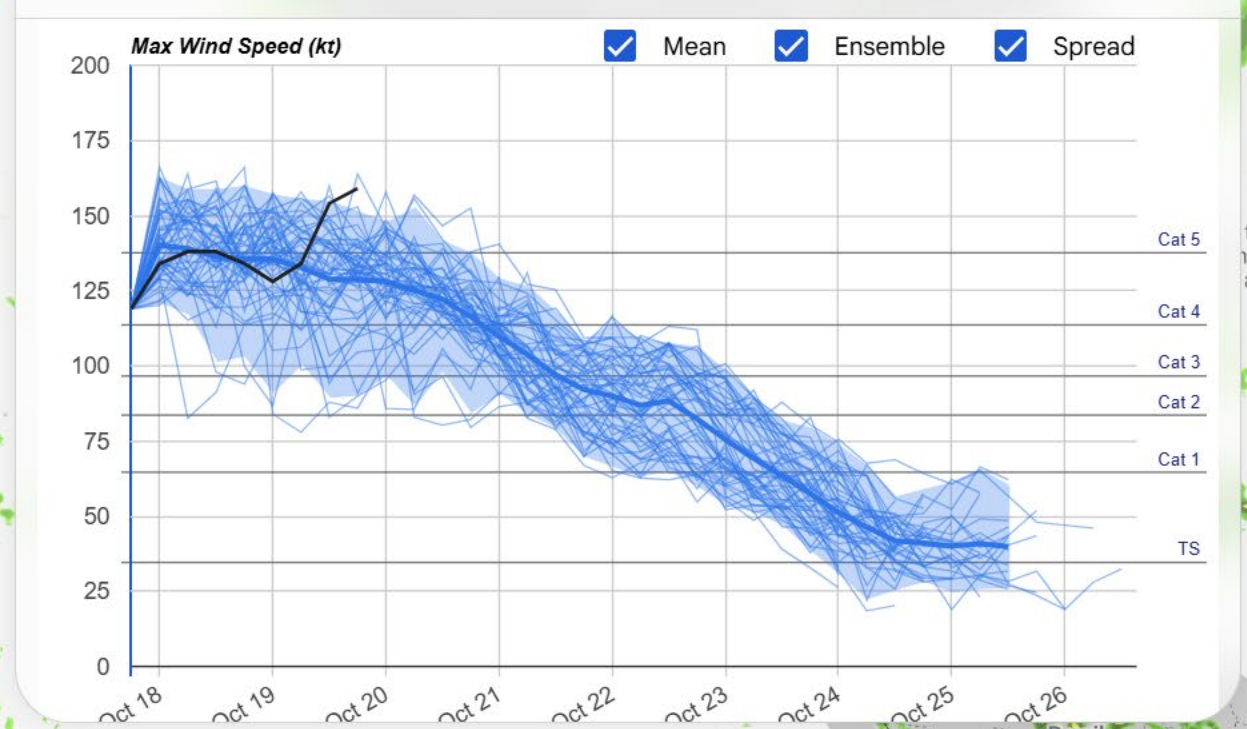
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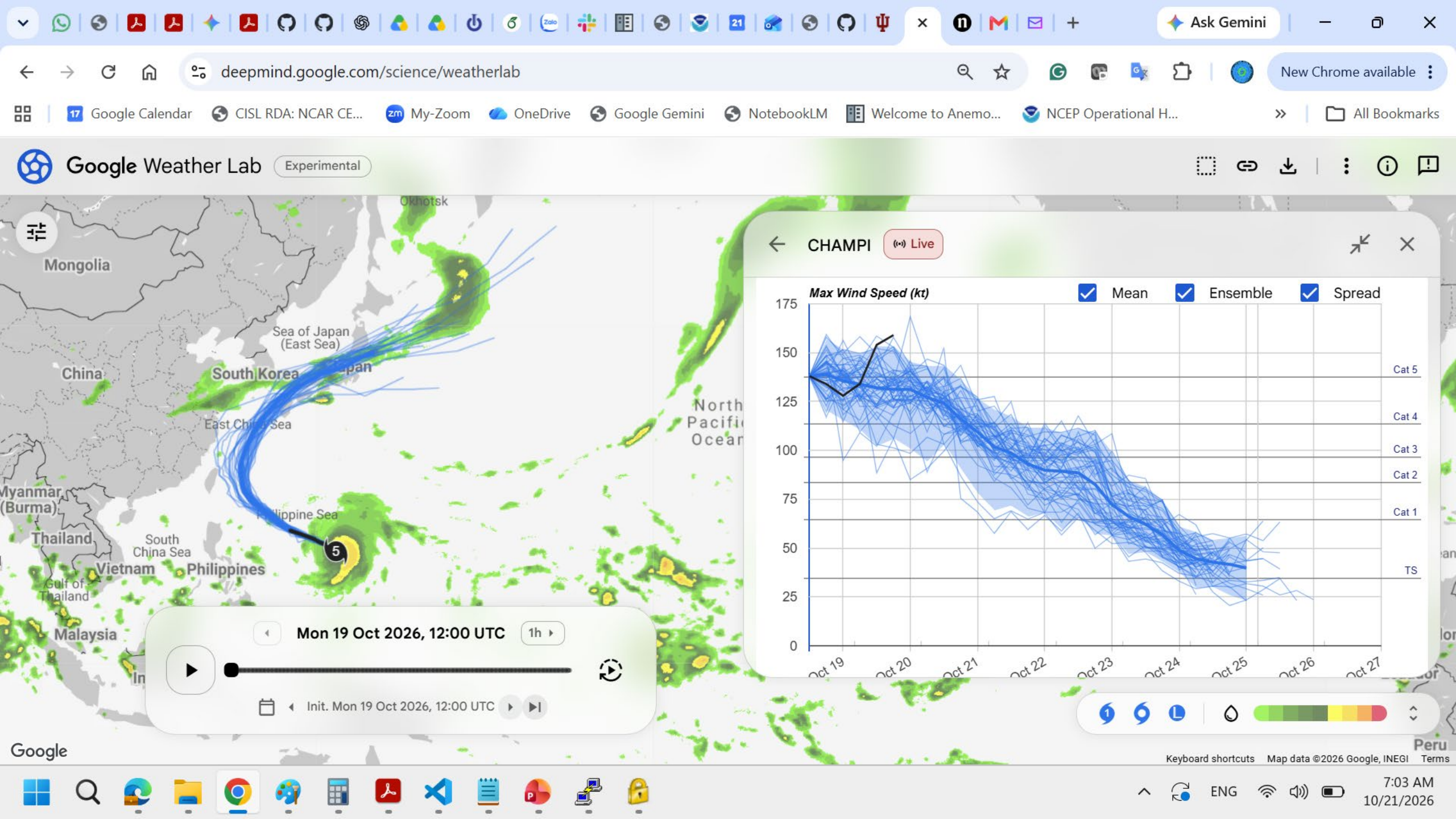
Sun 18 Oct 2026, 18:00 UTC 1h

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Bay of Bengal
South China Sea
Coral Sea
Brazil

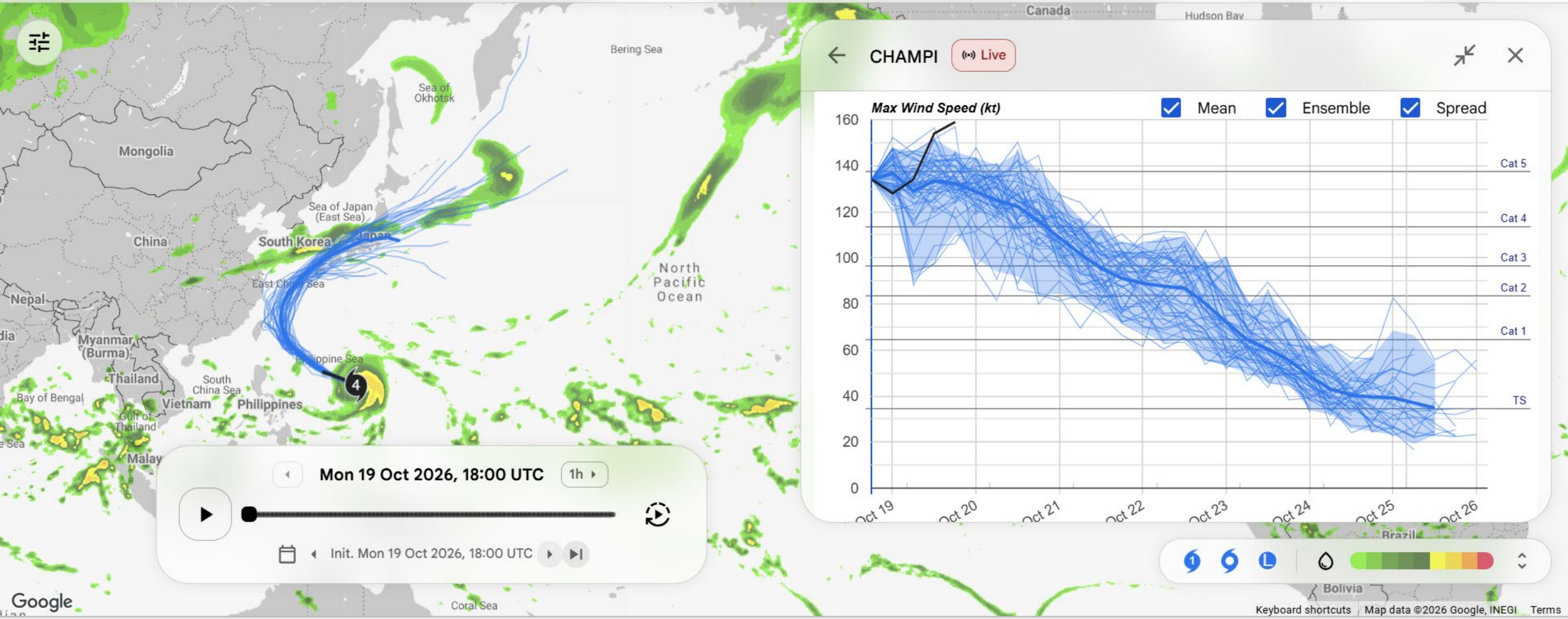
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Gulf of
Thailand
Myanmar
(Burma)
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Max Wind Speed (kt)

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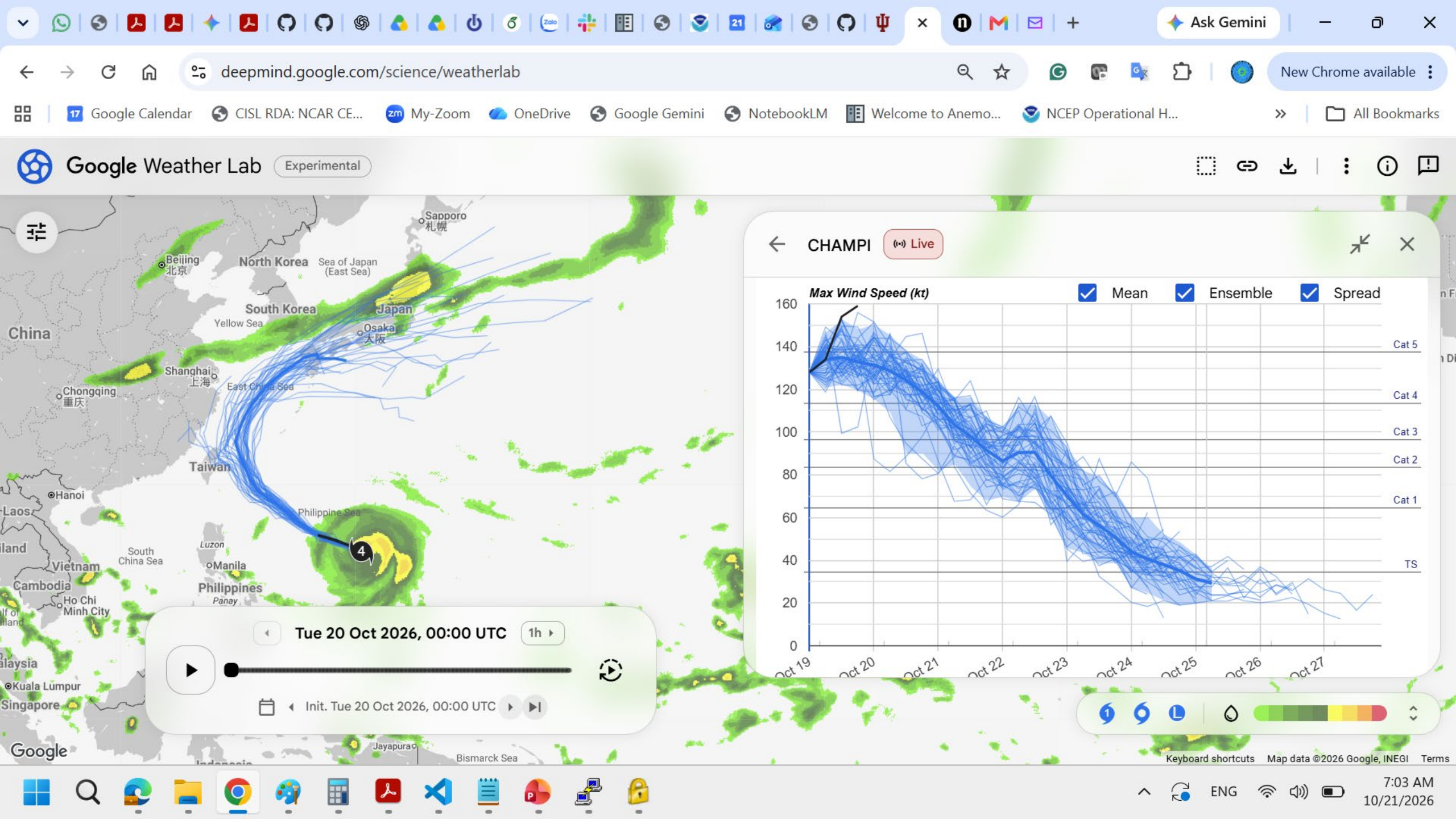
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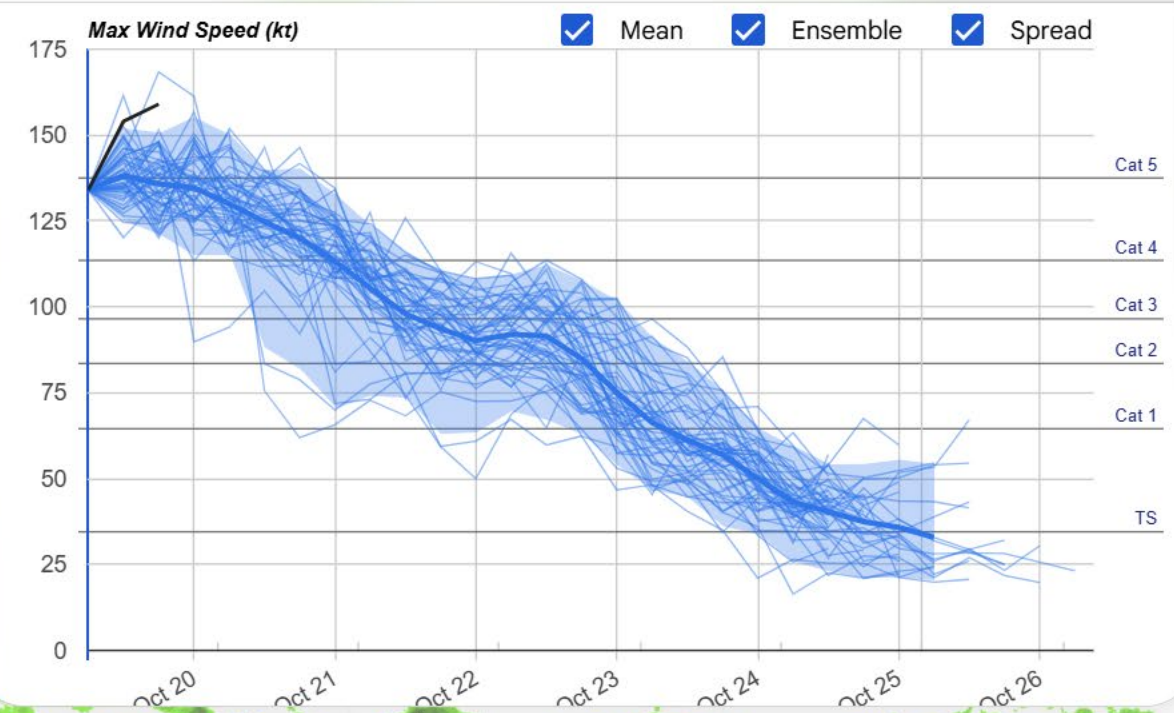
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Chongqing
重庆
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Taiwan
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Laos
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South
China Sea
Luzon
Manila
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Cambodia
Ho Chi
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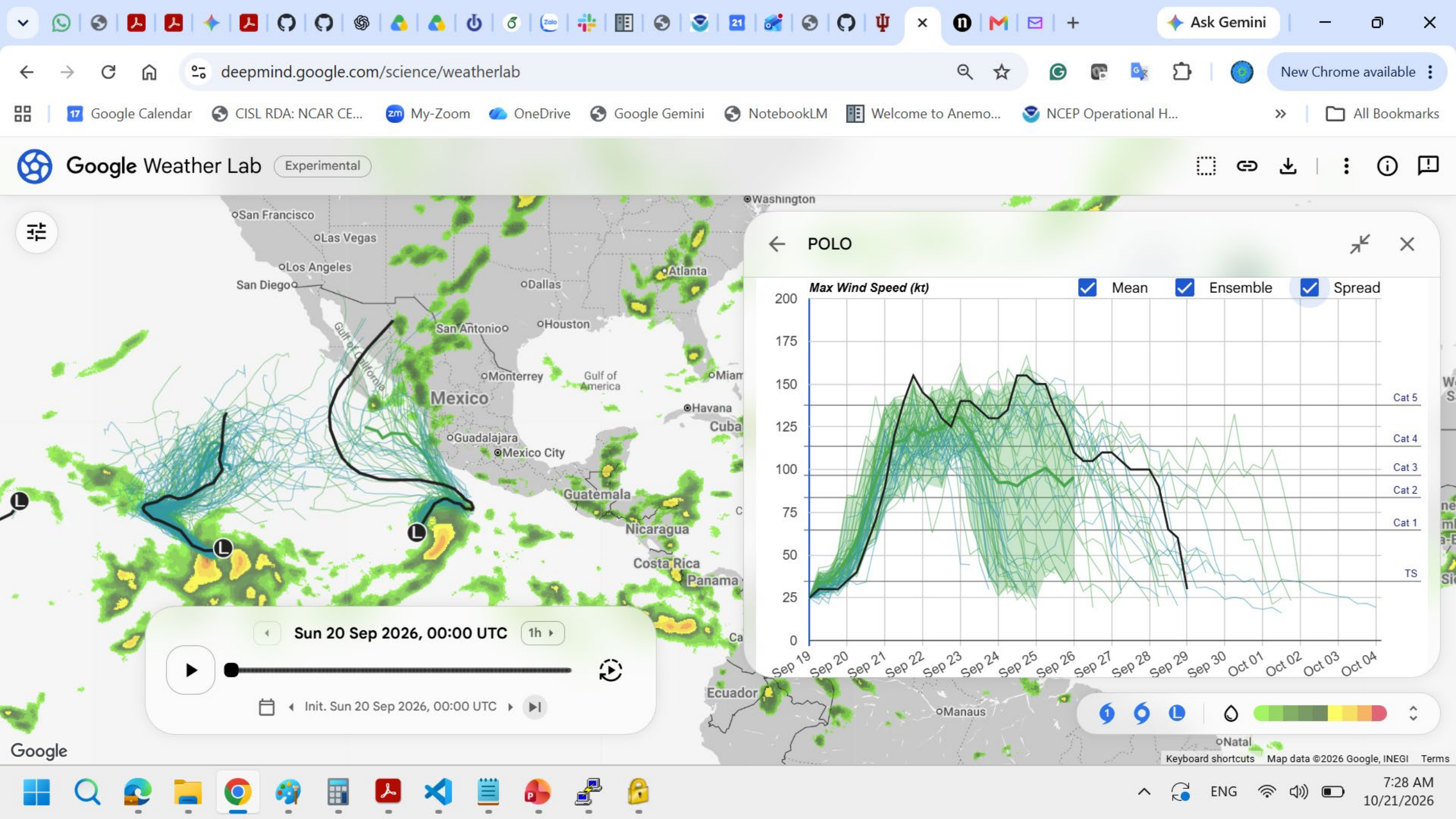
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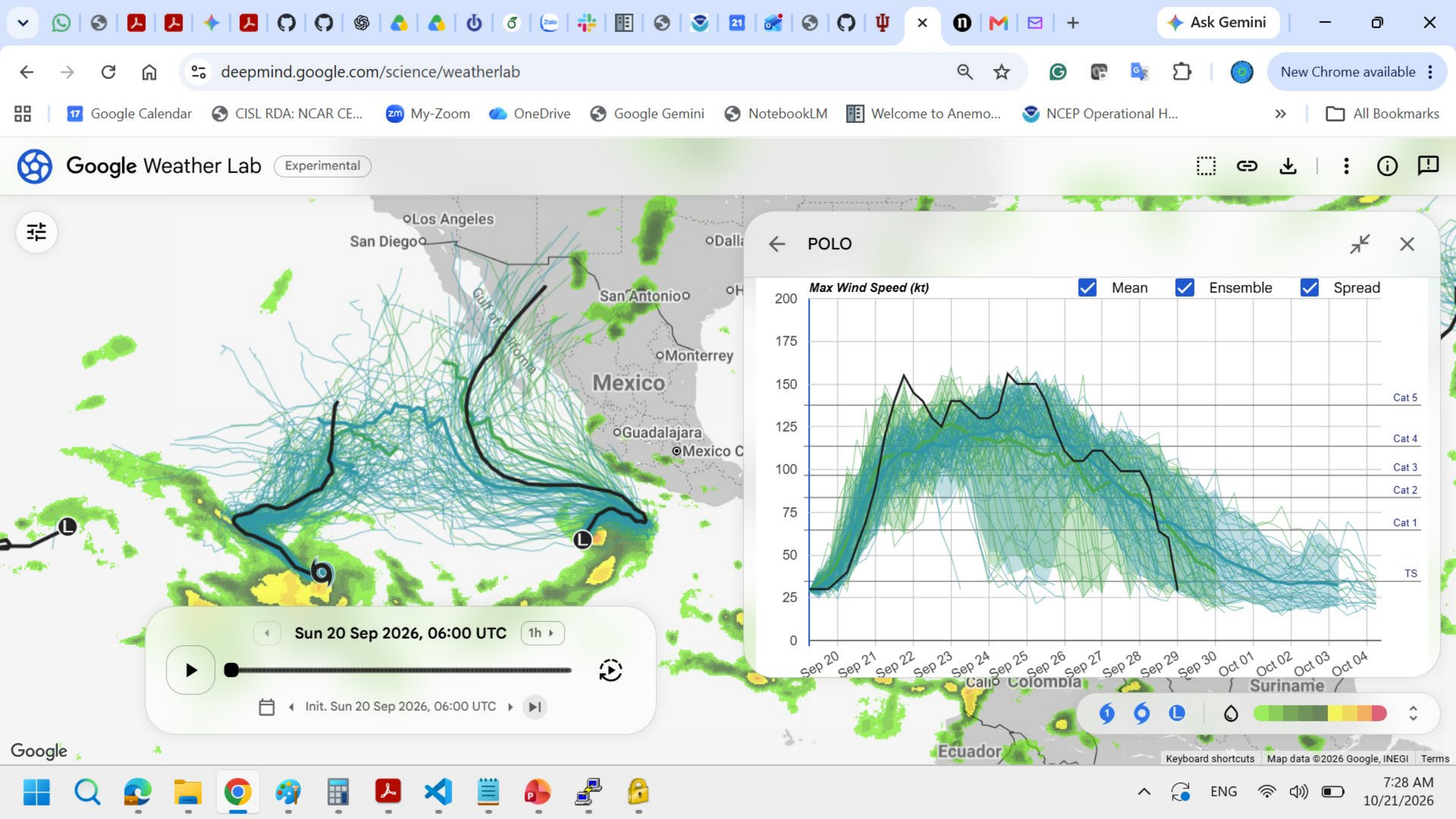
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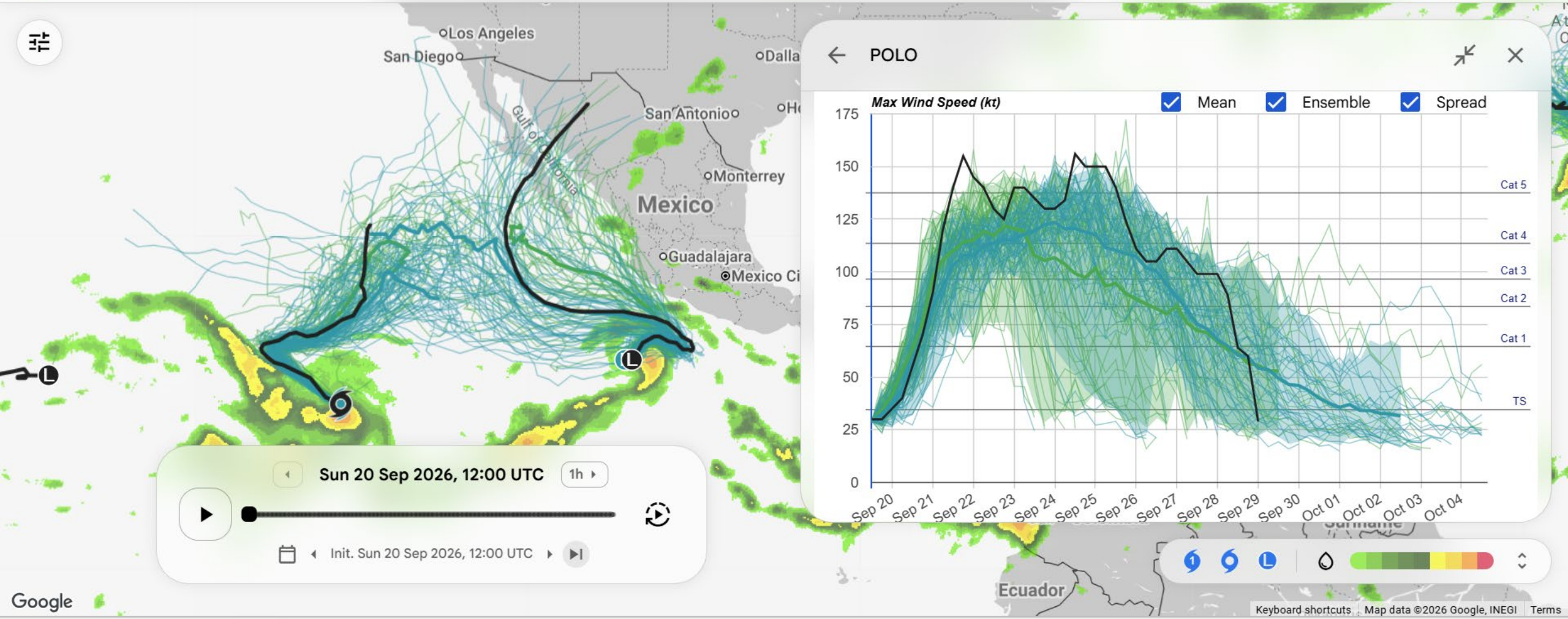
Hurricane Polo (Sep 2026)


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Guadalajara
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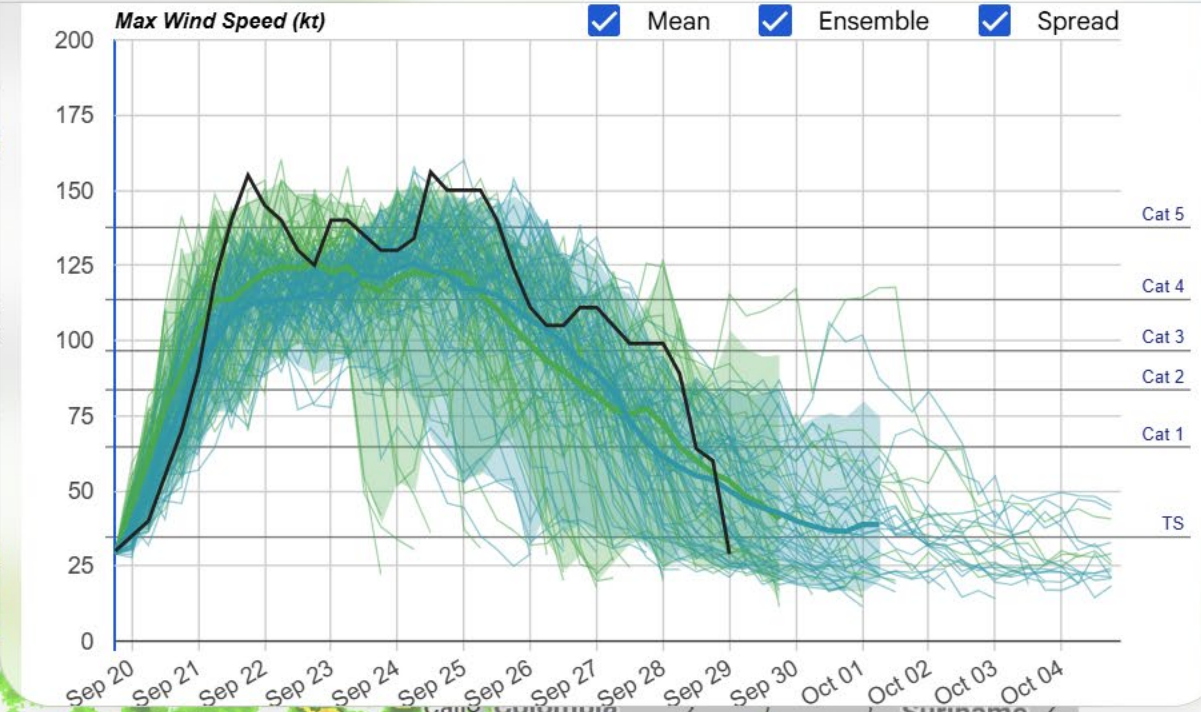
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Experimental

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Sun 20 Sep 2026, 18:00 UTC

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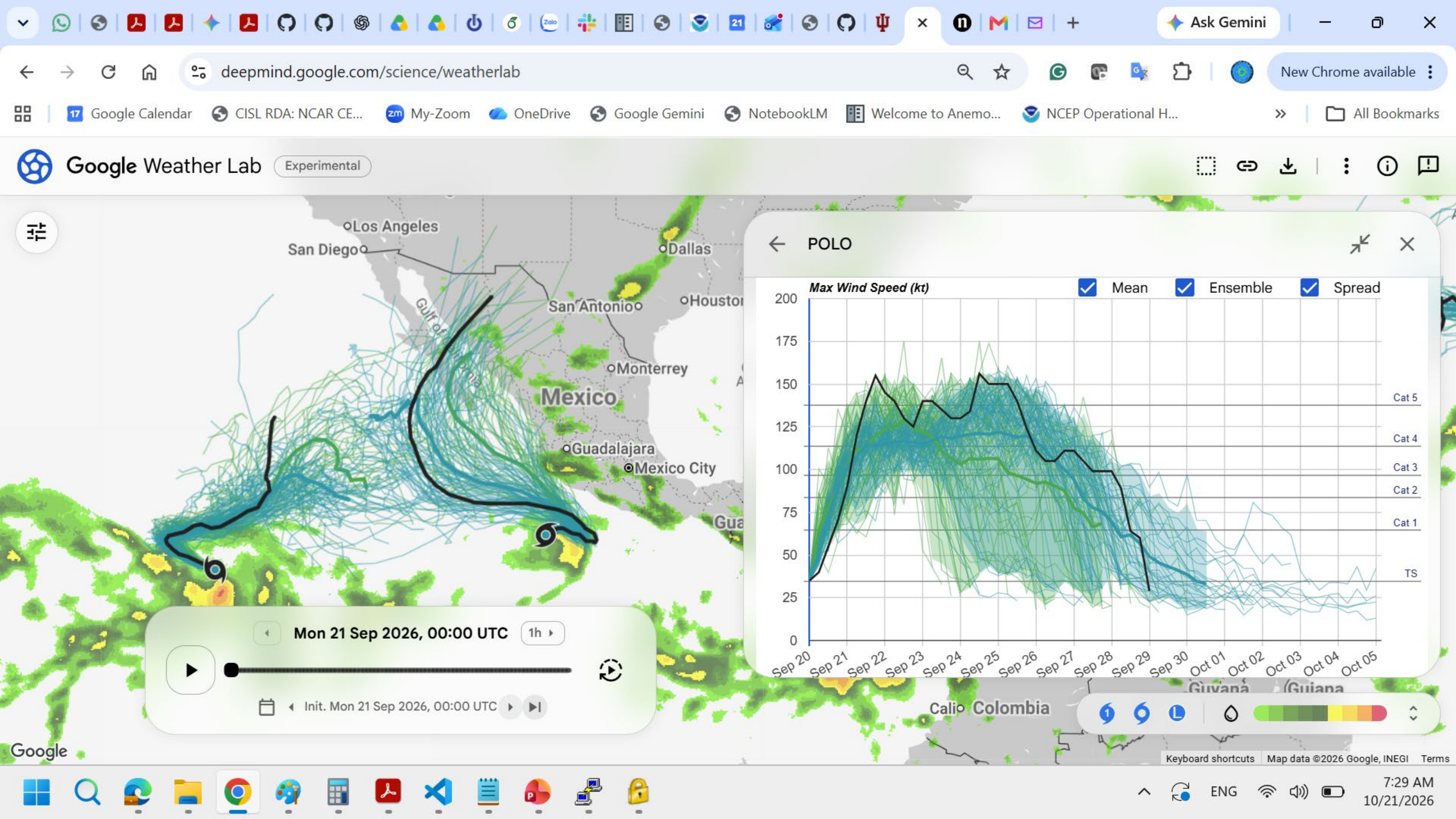
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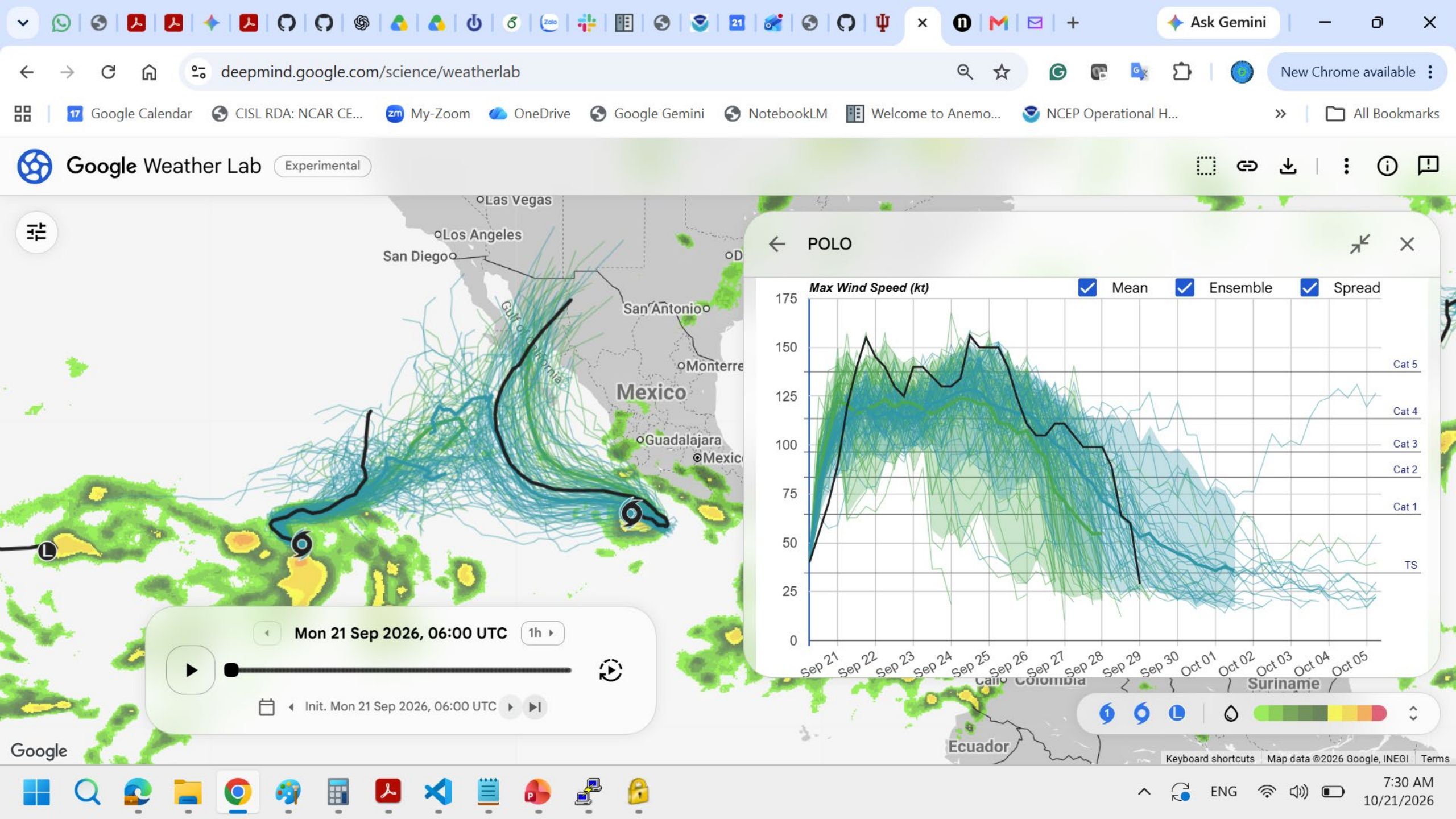
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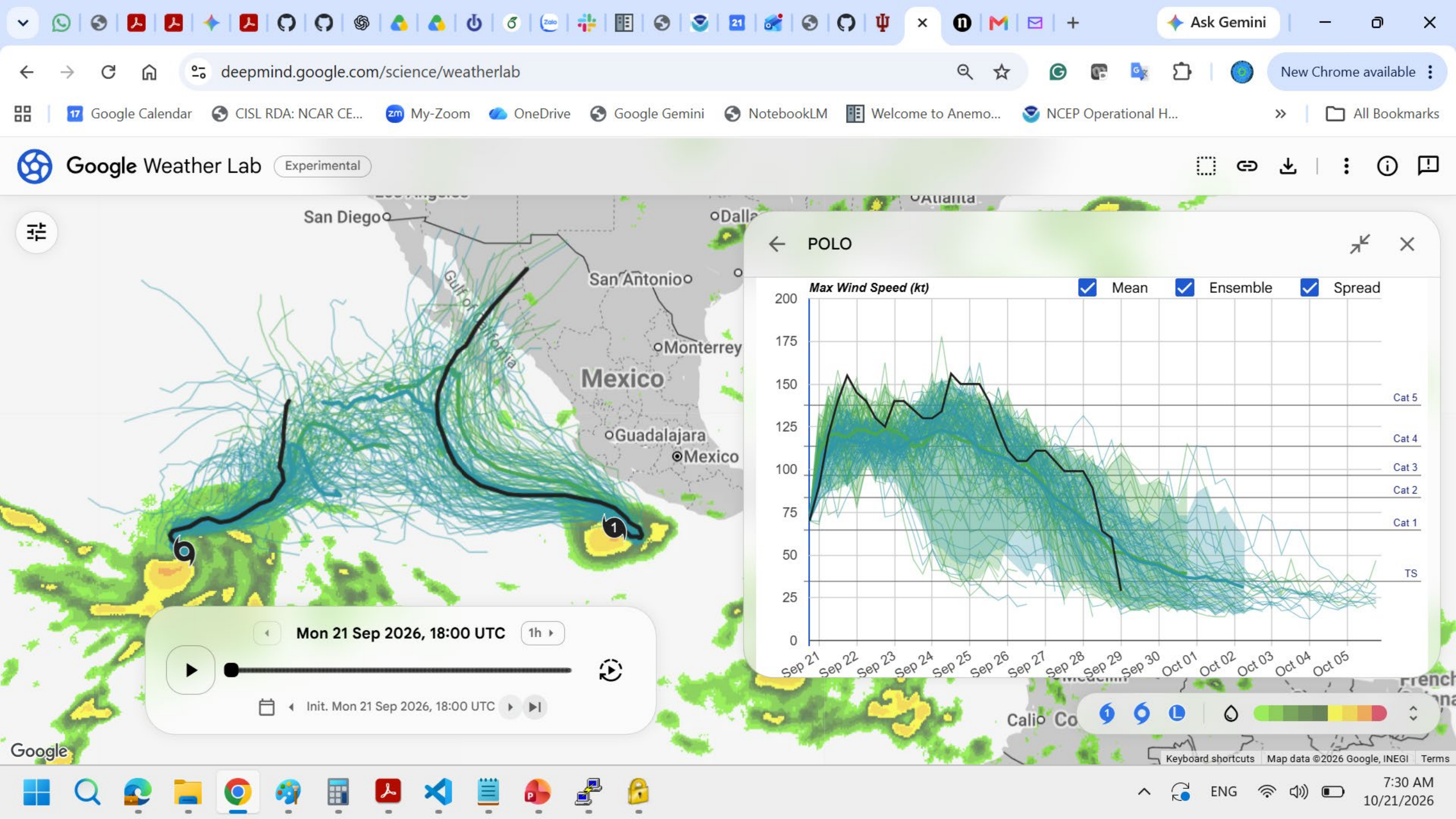
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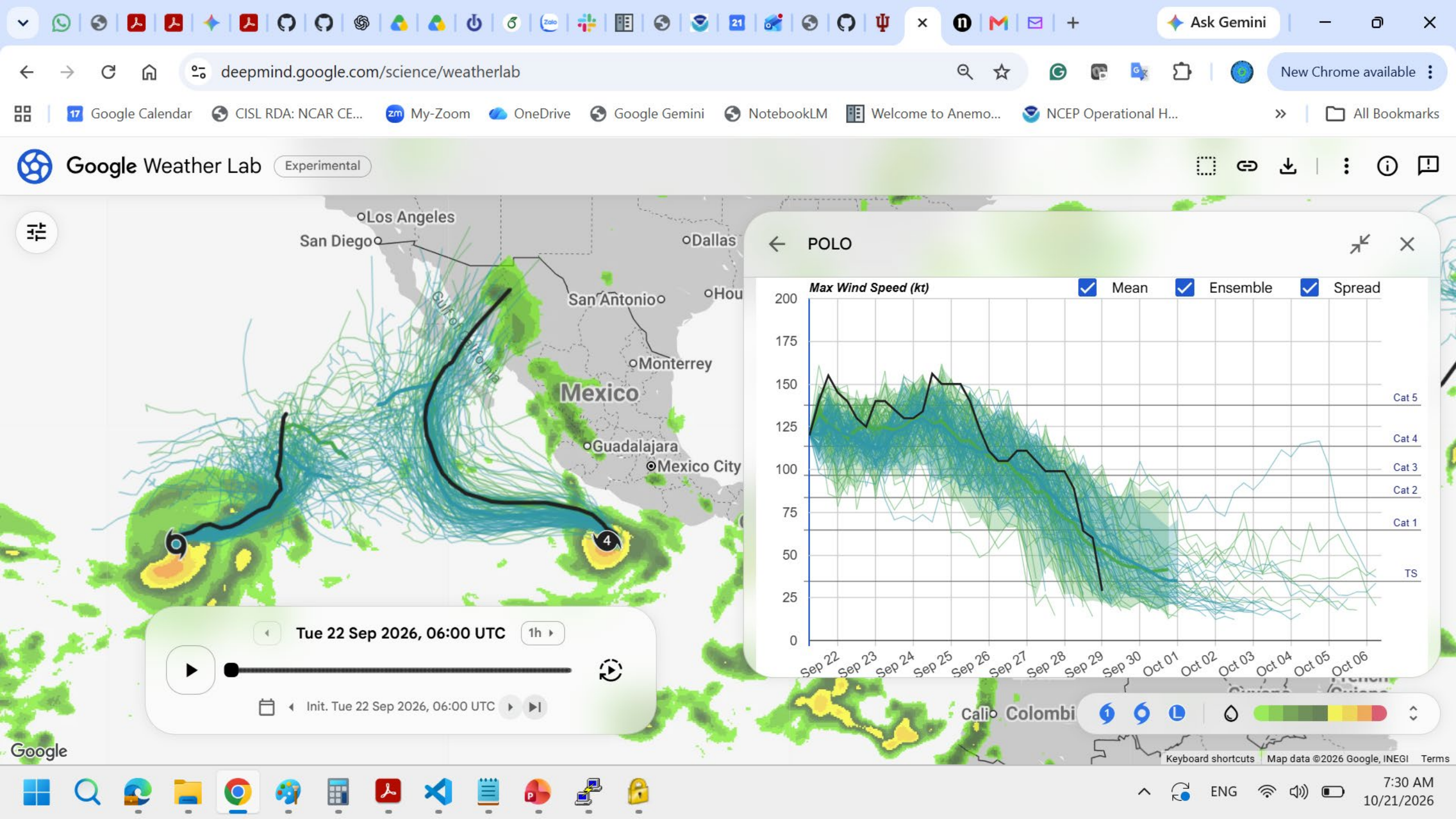
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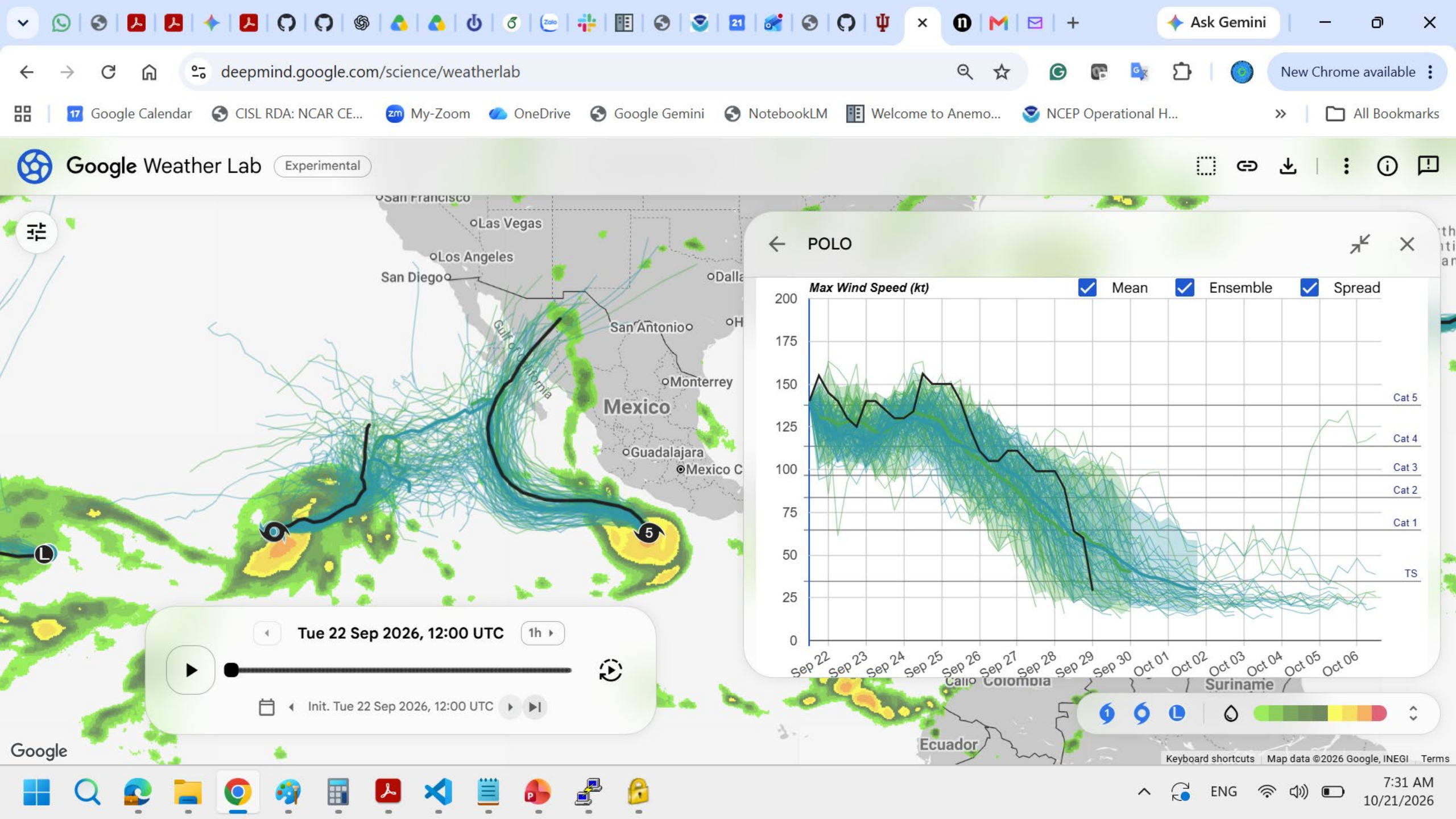
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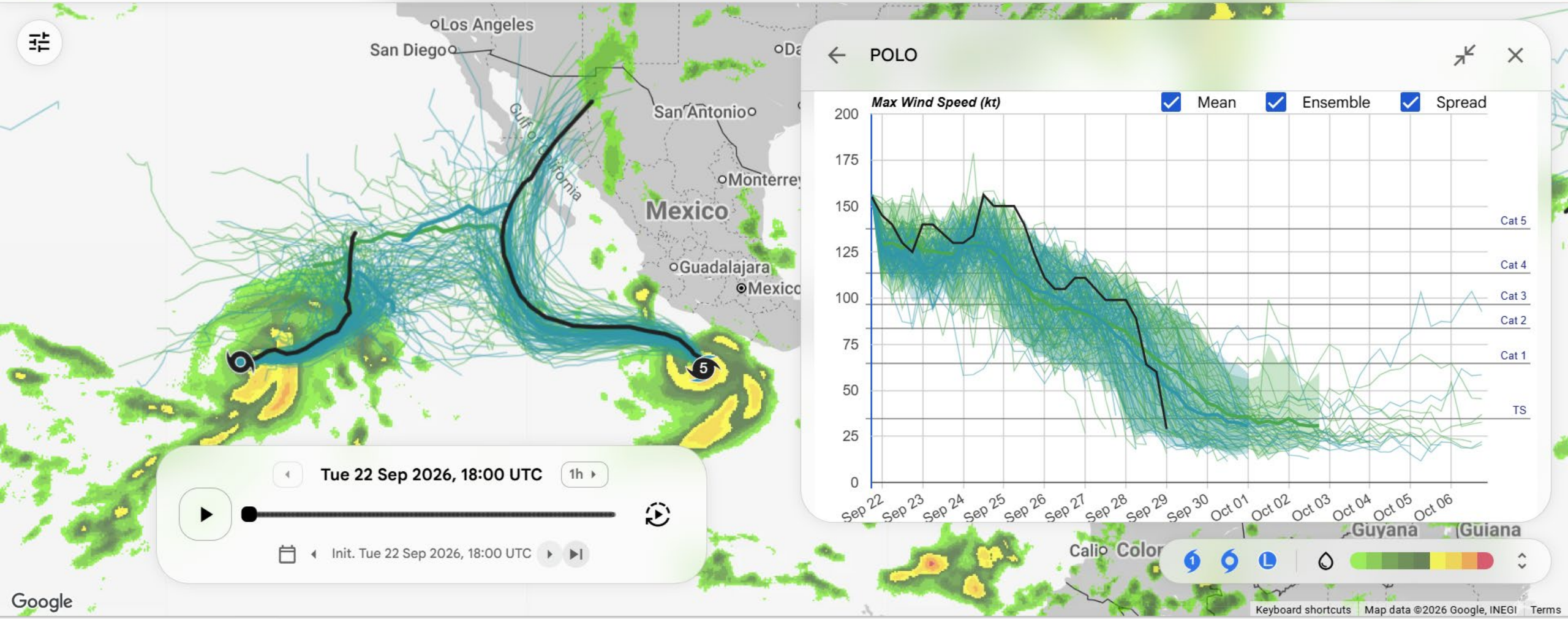

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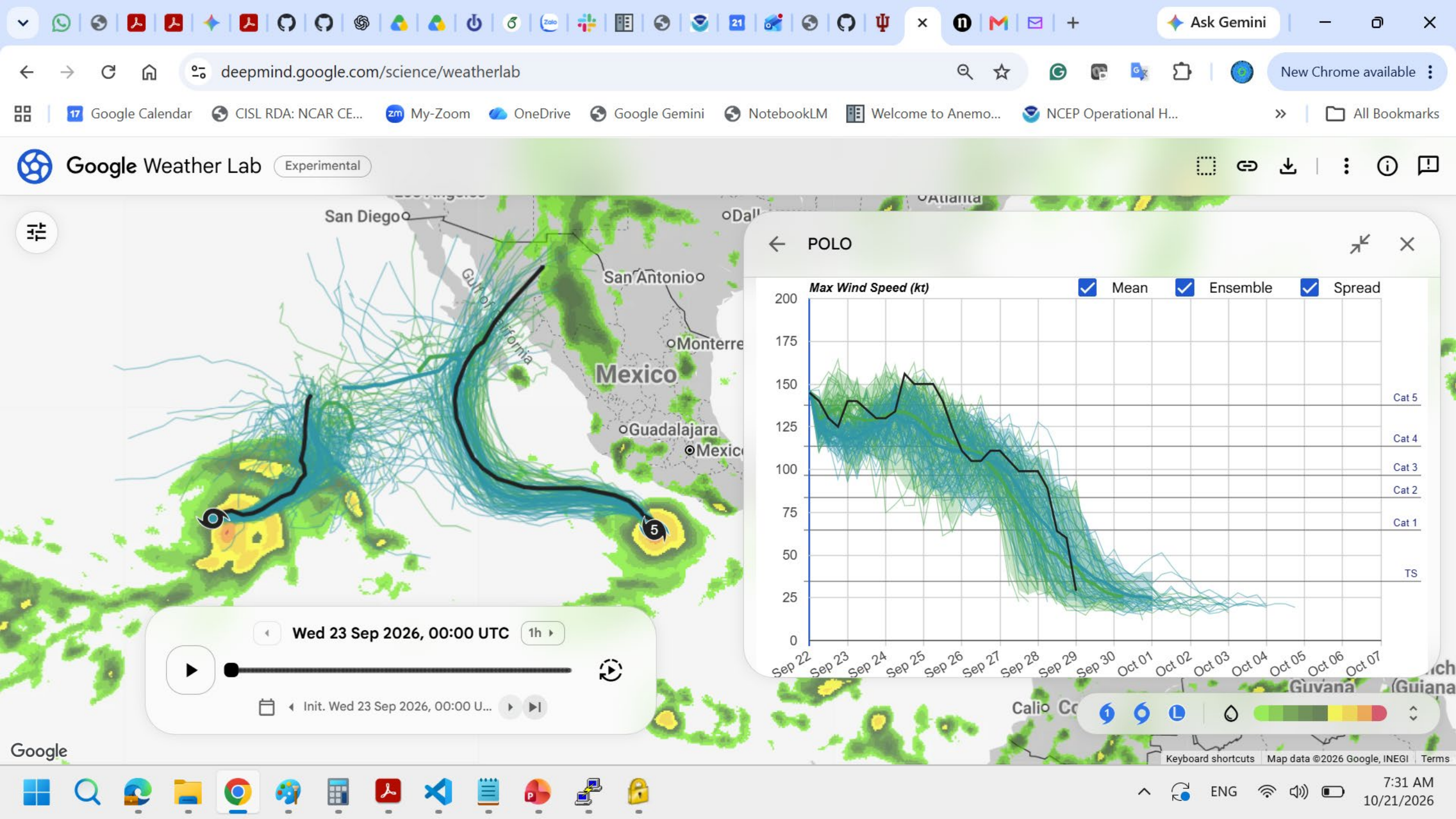

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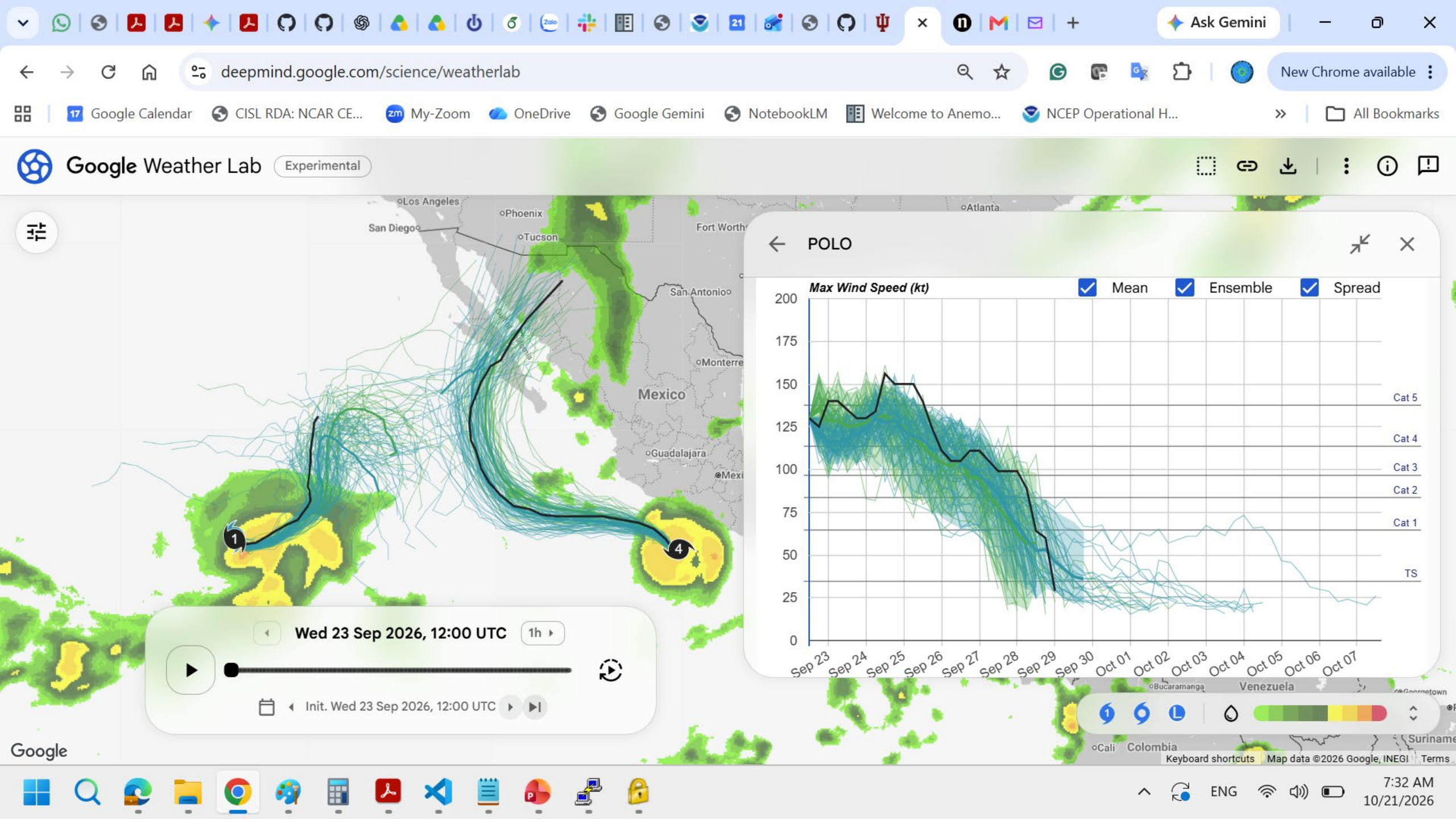
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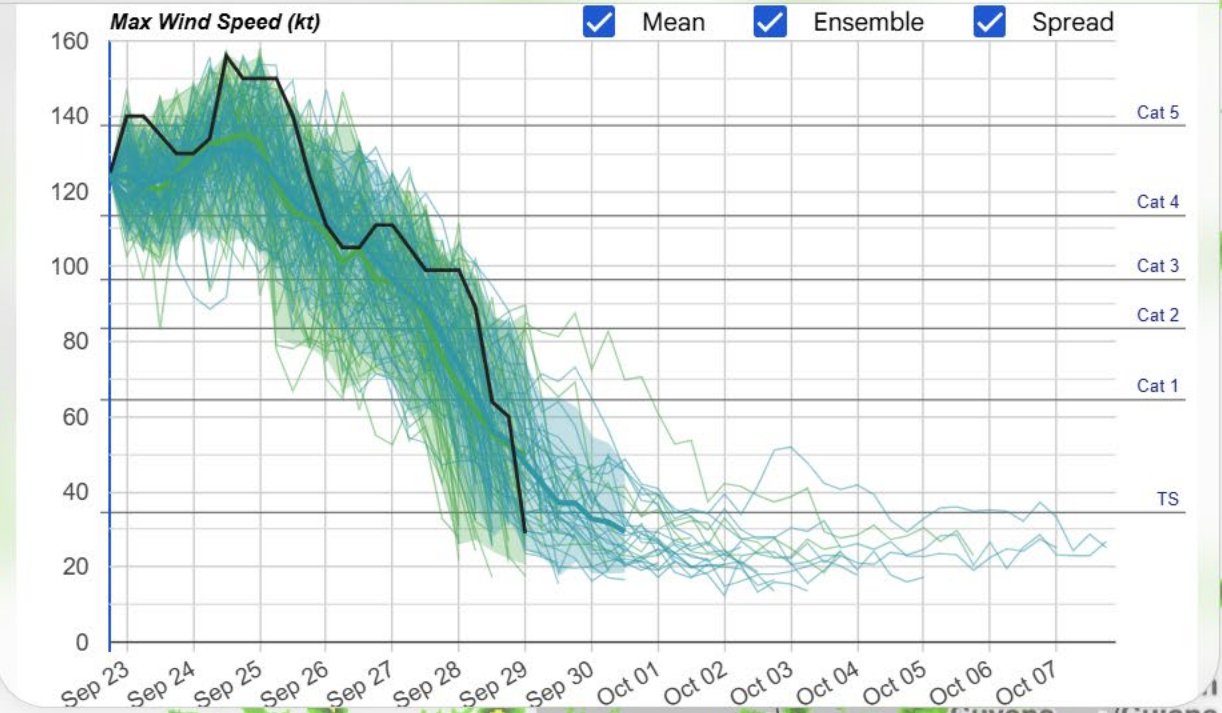
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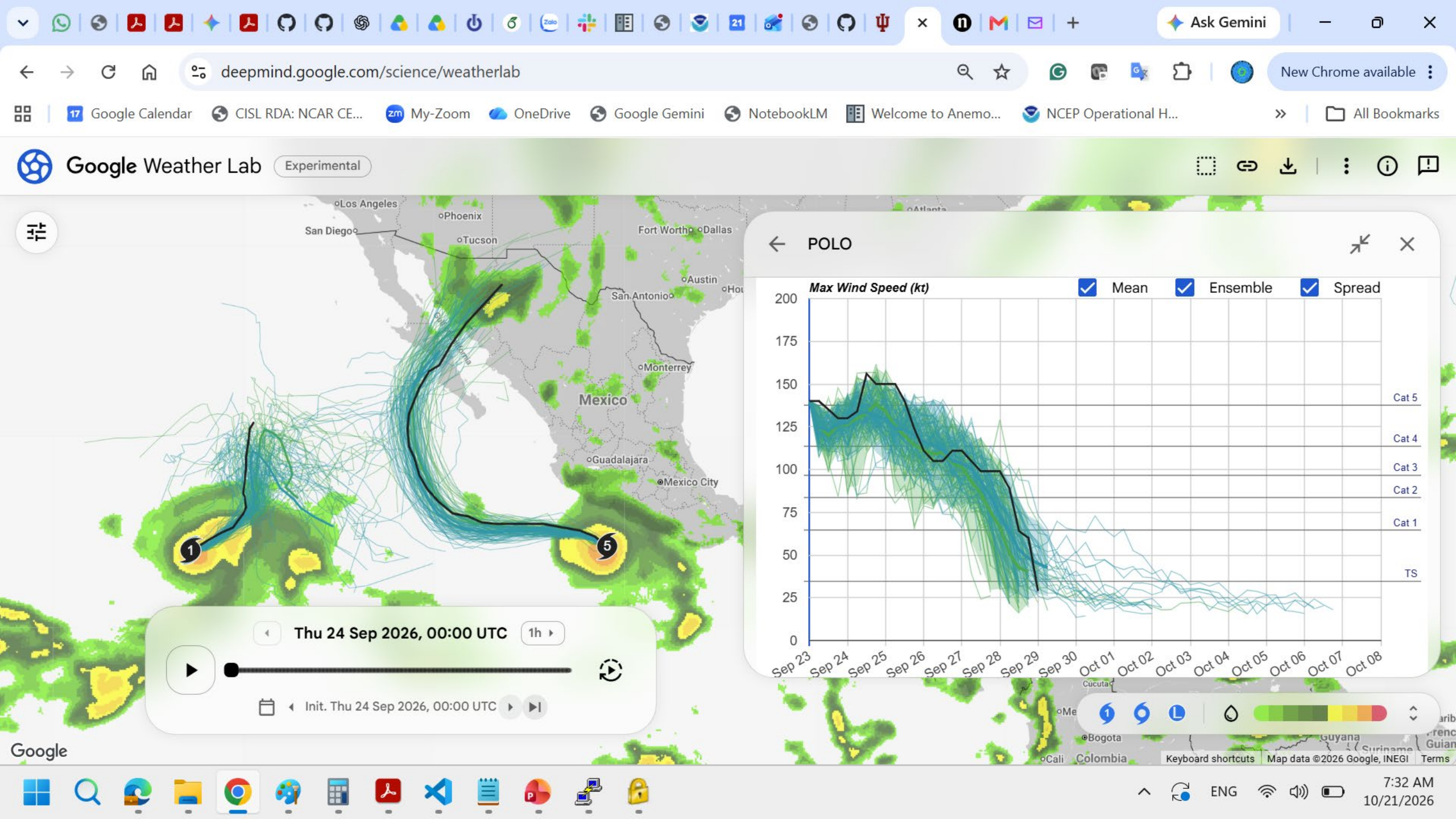
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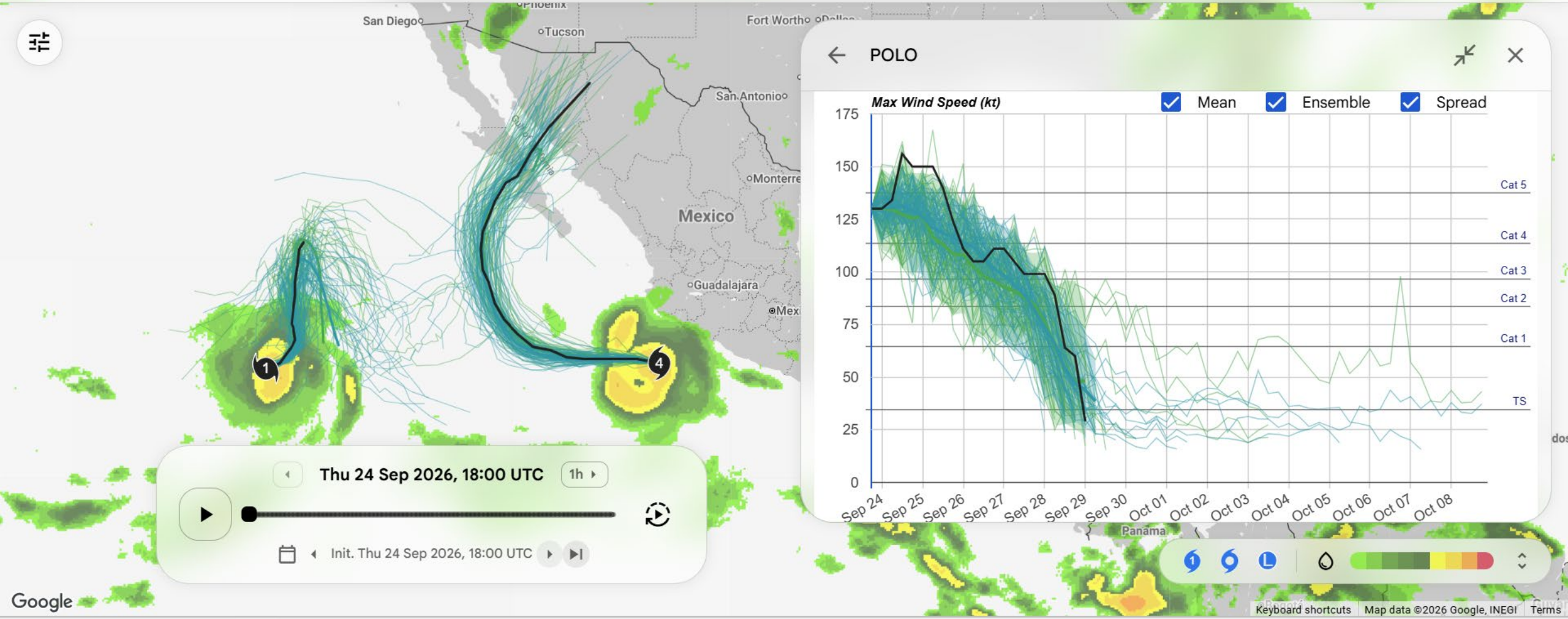
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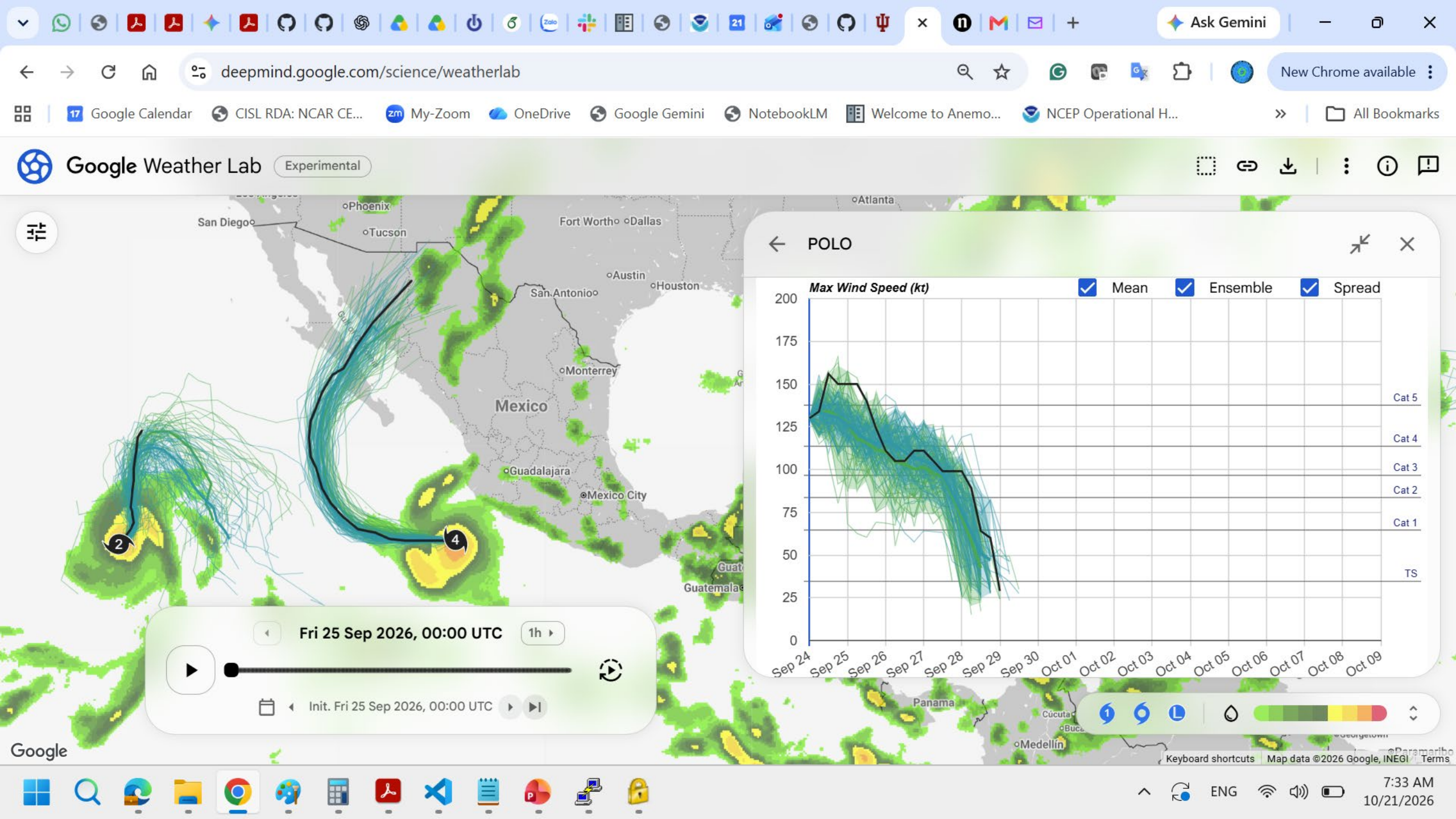

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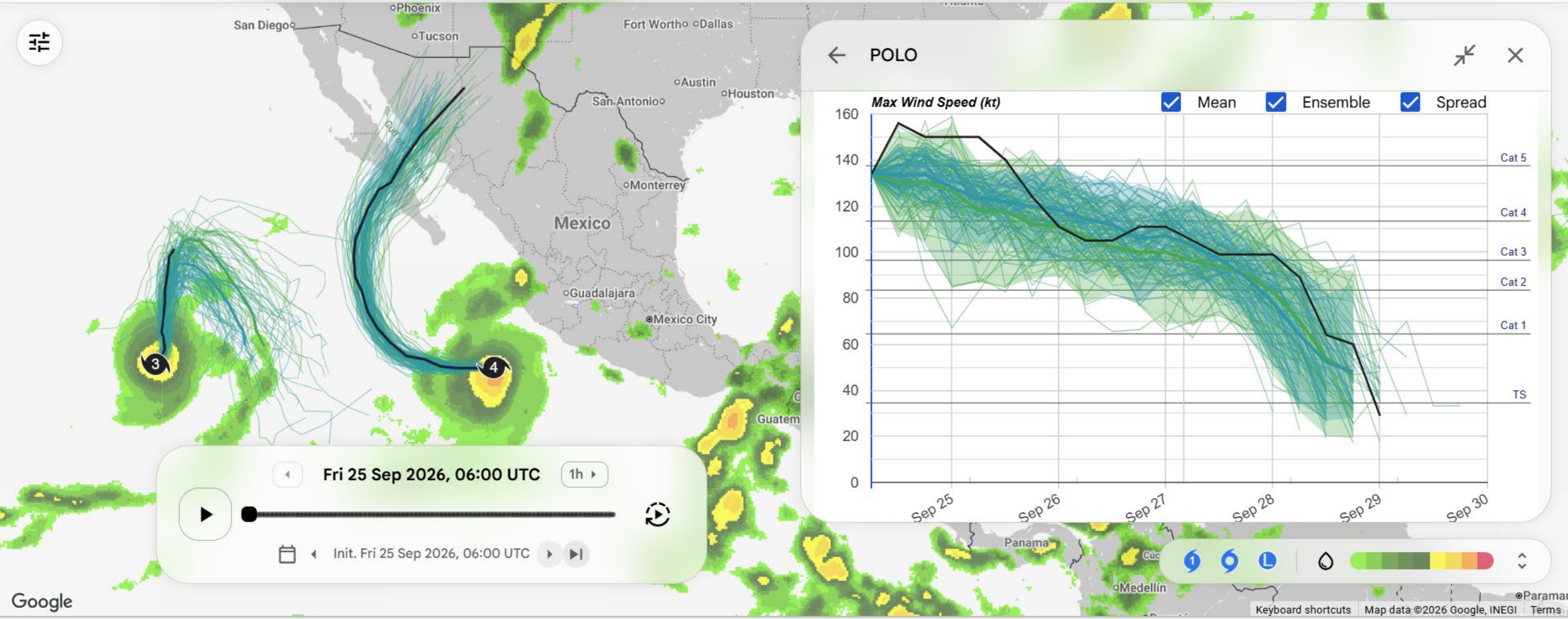
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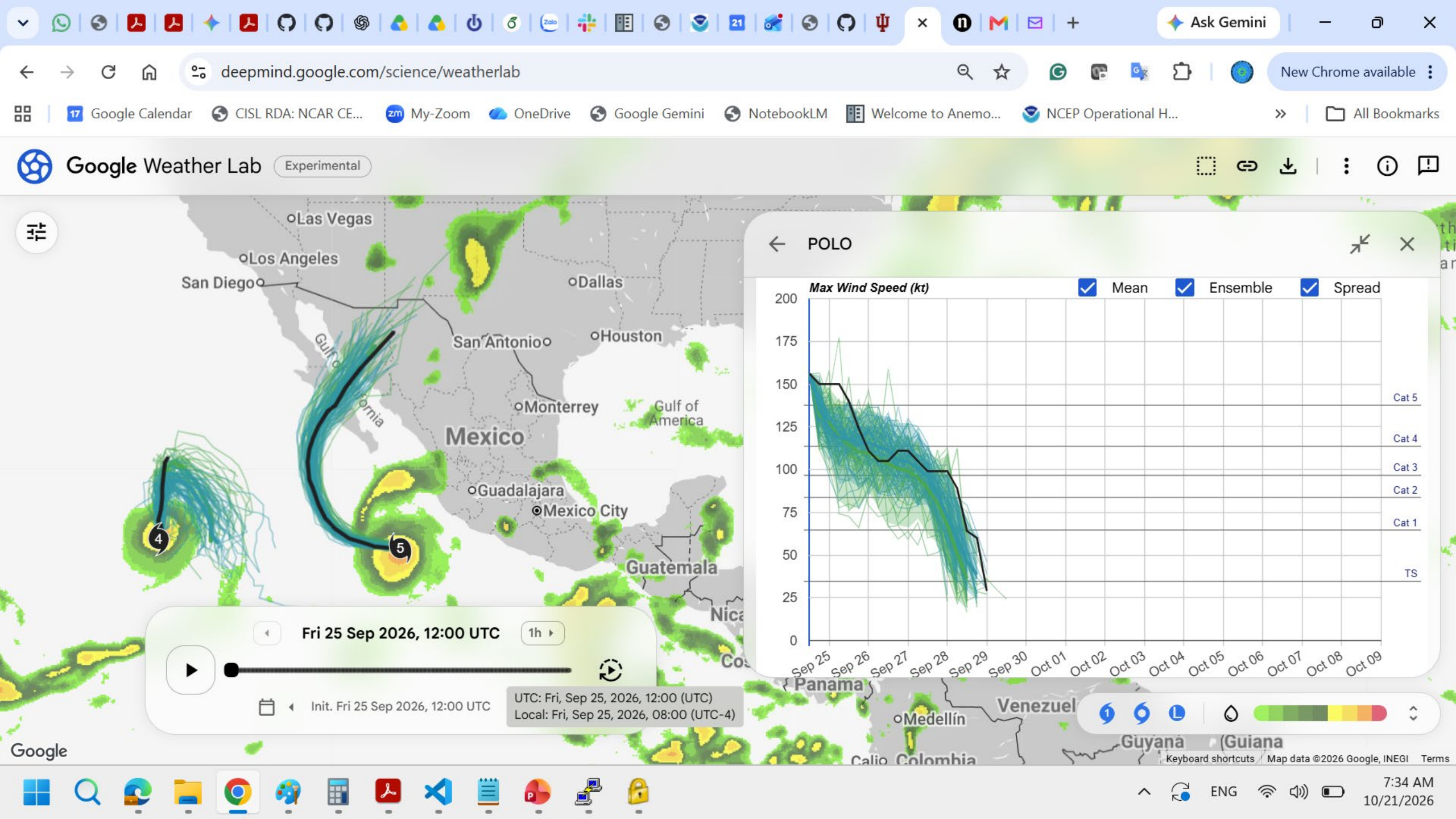
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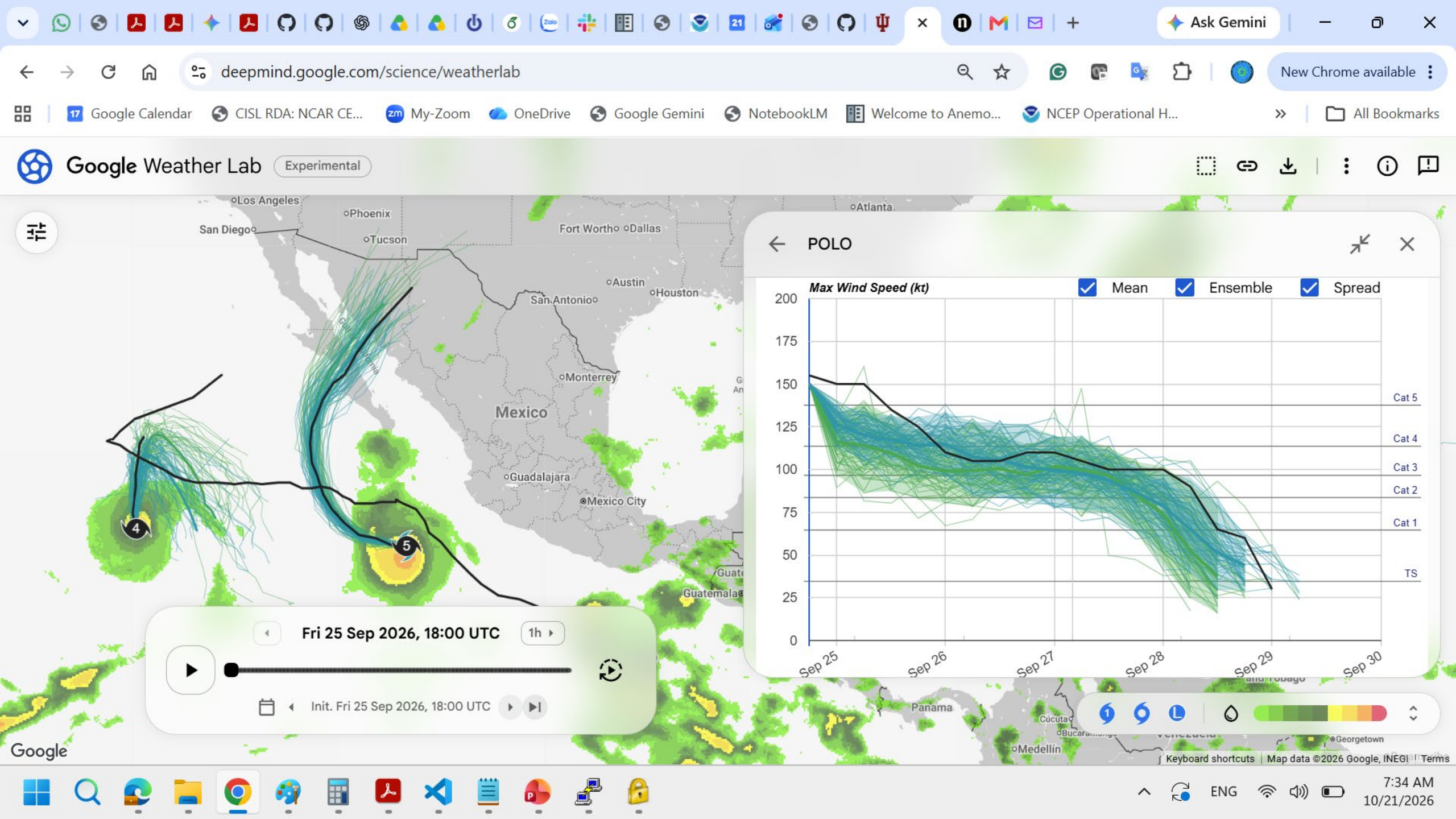


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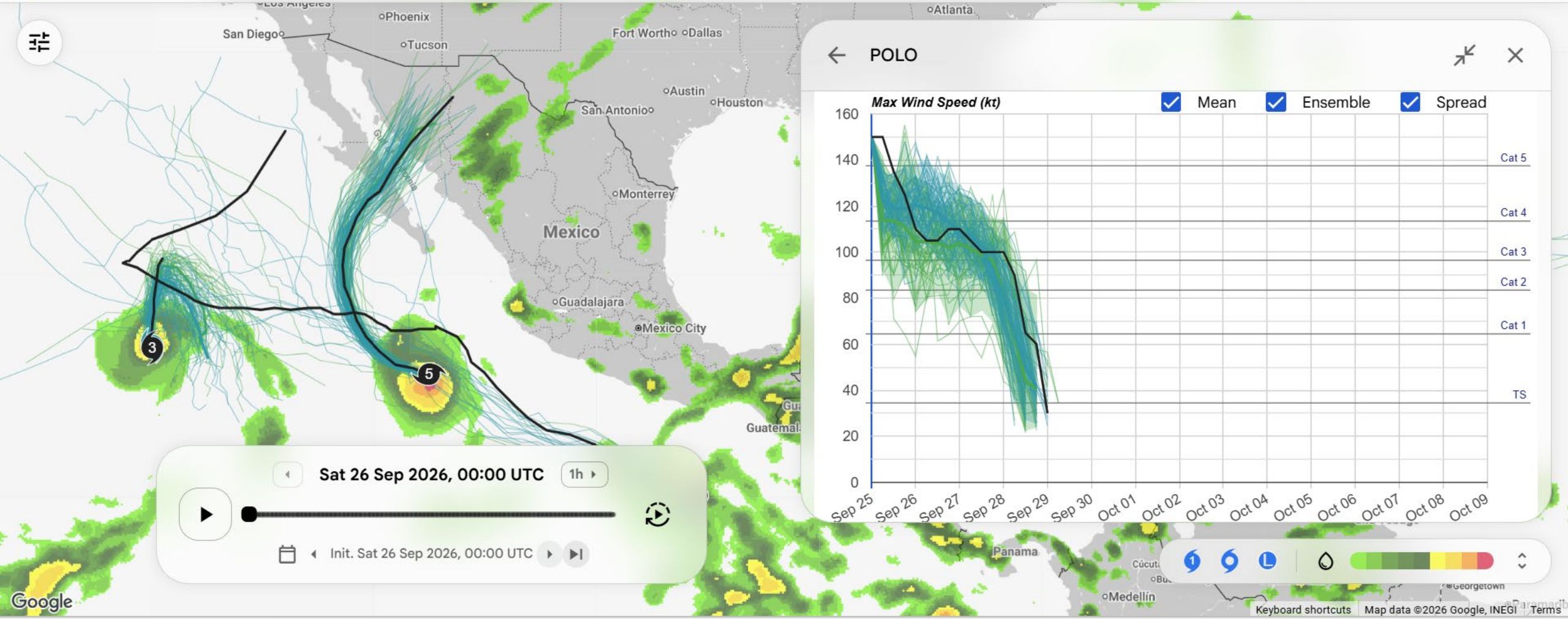
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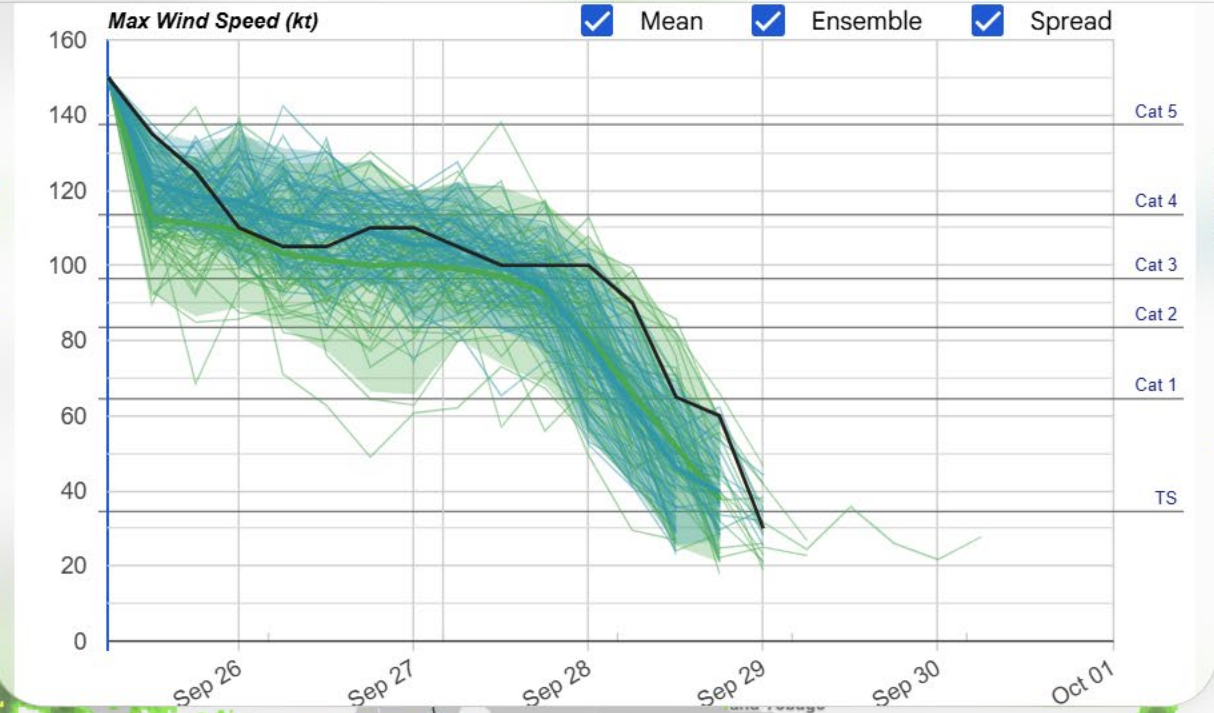
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Conclusion